Nowcasting and forecasting German house prices

Sunyoung Ji

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Abstract

This study evaluates the performance of Mixed-Data Sampling (MIDAS) and Bayesian Mixed-Frequency VAR (MF-VAR) approaches for nowcasting and forecasting German house prices. These mixed-frequency models provide timely updates of quarterly German House Price Index (HPI) growth by exploiting indicators sampled at higher (monthly and daily) frequencies and handling ragged-edge datasets caused by publication lags. Nowcasts and forecasts are produced recursively from single-indicator and pooled models within a pseudo-real-time framework, over two evaluation samples ending before and after the onset of the COVID-19 pandemic. The results show that mixed-frequency models improve accuracy most clearly under conditions of high economic uncertainty. The Bayesian MF-VAR delivers the most accurate nowcasts (horizon h = 0), while AR-MIDAS and U-MIDAS perform best at longer forecast horizons (h = 1 to 8 quarters). Pooling predictors improves the accuracy of MF-VAR, whereas it generally reduces the performance of MIDAS relative to its single-indicator models.

Keywords: Nowcasting; Forecasting; Mixed-frequency data; MIDAS; Mixed-frequency VAR (MF-VAR); Bayesian VAR; House prices; Real-time data

JEL Codes: C53, C32, C11, R31

1 Introduction

Understanding house prices in a timely manner is essential for stakeholders including policymakers, investors, real estate professionals and homeowners. During periods of economic uncertainty, precise and prompt forecasts become even more critical, supporting informed decisions on investment strategies, housing policies and overall economic stability. These insights support informed decision-making in areas such as investment strategies, housing policies and overall economic stability. This paper explores the application of nowcasting and forecasting models to German house prices with mixed-frequency models capable of handling different sampling frequencies of variables and the complexities generated by publication lags.

Researchers frequently encounter the following challenges when analyzing time series data. First, economic time series are sampled at different frequencies. Macroeconomic variables are primarily published monthly or quarterly, depending on the statistical agencies, while financial variables can be observed daily or even in real-time. Second, a certain amount of time is required for data to be collected and published, as the process involves several steps, including data processing, validation, analysis, the application of statistical techniques (e.g., seasonal adjustments and weighting) and final review and approval. These delays in publication result in datasets with a ragged-edge structure, where different time series within the same dataset have varying lengths of available data. This mismatch in sampling frequencies and the ragged-edge structure make the dataset imbalanced and complicate the application of traditional time series models. The simplest solutions to addressing these issues are pre-filtering and trimming the end of the dataset. Pre-filtering involves matching left- and right-hand variables at the same frequency. For example, if quarterly HPI data and monthly macroeconomic variables are used, researchers can aggregate the monthly data to a quarterly frequency. This can be done by calculating the quarterly average of the monthly data or using a representative value, such as the data from the last month of each quarter. In addition, to cope with ragged edges in the dataset, a common practice is to truncate the ends of the series to ensure that all series within the dataset have the same length. These methods artificially balance the dataset, ensuring it is suitable for analysis using standard time series models. However, this can

destroy potentially valuable information, such as the most recent observations from high-frequency variables, or lead to misspecification. These limitations have increased the need to directly model unbalanced datasets, leading to the development of mixed-frequency models. Mixed-frequency models directly address the imbalancedness of datasets, enabling the use of the latest data for timely nowcasts and forecasts. Hence, these models are particularly advantageous for generating more accurate predictions during periods of economic turbulence, such as the COVID-19 pandemic, which significantly disrupted global markets. In this paper, I focus specifically on state-of-the-art mixed-frequency models: Mixed Data Sampling (MIDAS) and Mixed-Frequency Vector Autoregression (MF-VAR).

MIDAS regression, introduced by Ghysels, Santa-Clara and Valkanov (2004), involves a dependent variable and explanatory variables sampled at different frequencies. In linear models, using higher-frequency predictors can lead to a significant increase in the number of parameters to be estimated. To prevent the proliferation that arises from this increase, MIDAS employs parsimonious distributed lag polynomials for high-frequency explanatory variables. Clements and Galvão (2008) expand MIDAS by incorporating AR dynamics, referred to as AR-MIDAS. Initially used for financial applications, MIDAS has since been applied to macroeconomic time series, such as quarterly GDP and monthly predictors (see, for example, Clements and Galvão, 2008, 2009; Marcellino and Schumacher, 2010). However, when considering monthly and quarterly variables, the difference in their sampling frequencies is relatively small compared to more disparate frequencies, such as those between daily and monthly variables. In such cases, the issue of parameter proliferation may not be a concern, thus, restriction on lag polynomials may not be required. Hence, Foroni, Marcellino and Schumacher (2015) develop a MIDAS model with an unrestricted linear lag polynomial, demonstrating that unrestricted MIDAS (U-MIDAS) performs better than MIDAS under these conditions. In the remainder of this paper, 'MIDAS models' refer to the basic MIDAS, AR-MIDAS and U-MIDAS.

While MIDAS is a single-equation approach, MF-VAR is a system approach that jointly explains variables sampled at different frequencies. To manage different sampling frequencies, MF-VAR

is represented as a state-space model that evolves at a monthly frequency. In this paper, I employ the Bayesian framework for MF-VAR. Within this framework, I allow the hyperparameters for the dynamic components to change over time and incorporate the steady-state (unconditional mean) prior proposed by Villani (2009) into the MF-VAR, following Ankargren, Unosson and Yang (2018). Given the observed data, I use the Gibbs sampler to estimate parameters and latent data.

The performance ranking of MIDAS models and MF-VAR depends on the underlying data-generating process (DGP). Thus, a comparative analysis of their predictive accuracy should be conducted within the context of specific empirical applications. In this study, I focus on nowcasting and forecasting the growth of the German HPI and compare the accuracy of nowcasts and forecasts obtained from these mixed-frequency models. For different forecast horizons, the comparison is based on the mean squared forecasting error (MSFE) relative to a naive benchmark, which is an AR model using only quarterly series. The analysis is conducted recursively with an expanding estimation sample over time in a pseudo-real-time manner. To determine whether the economic crisis caused by the COVID-19 pandemic impacted the results, two evaluation samples are used: one ending before the crisis and one after. This procedure is applied to models for both single predictors and the combined dataset including all predictors selected for this study. To investigate the performance of the combined predictor models relative to the single predictor models, I calculate the percentiles of the MSFEs from the combined predictor models within the empirical distribution of all MSFEs from the single predictor models.

The main findings of this study highlight the superior performance of mixed-frequency models compared to the benchmark, producing more accurate nowcasts and forecasts during periods of economic instability. Specifically, the MF-VAR model demonstrates exceptional nowcasting accuracy, outperforming the MIDAS models. In terms of forecast horizons ranging from one to eight quarters ahead, AR-MIDAS produces the best forecasts. However, the results of the Diebold-Mariano test indicate that the differences in nowcast and forecast accuracy between AR-MIDAS and U-MIDAS are not statistically significant in general. Given the computational simplicity of U-MIDAS, it clearly stands out as a strong alternative to AR-MIDAS. Additionally, the results for the combined

predictor models indicate that incorporating information from multiple predictors improves accuracy in MF-VAR, particularly for two to four-quarter-ahead forecasts. However, in MIDAS models, the accuracy of nowcasts and forecasts decreases compared to single predictor models. These results emphasize the importance of selecting appropriate mixed-frequency models based on the specific forecast horizon, the set of predictors and the economic conditions.

This study makes contributions in two areas. On the methodological front, it provides both theoretical and empirical insights into the relative advantages of MIDAS models and MF-VAR. This comparative analysis serves as a valuable resource for future research, helping in the selection of the most appropriate model for specific research setups. Notable comparative studies in the literature include works by Kuzin, Marcellino and Schumacher (2011) and Foroni and Marcellino (2014), who compared the performance of MIDAS, AR-MIDAS and frequentist MF-VAR models for nowcasting GDP. Building on their research, this study extends the analysis by incorporating subsequent advancements, specifically by integrating U-MIDAS into the set of MIDAS models for comparative evaluation. This integration assesses whether U-MIDAS, which does not impose restrictions on lag polynomials, offers enhanced flexibility and improved forecasting accuracy. Furthermore, this study enhances MF-VAR by equipping it with a Bayesian framework. In a frequentist context, classical MF-VAR estimates parameters without restrictions, risking overfitting. In contrast, Bayesian VAR combines the researcher's prior beliefs with data using informative prior distributions, thereby reducing estimation uncertainty. Unlike the two aforementioned studies, this study expands the forecast horizon from zero to eight quarters, thus addressing both nowcasting and longer-term forecasting. Moreover, this study evaluates the performance of mixed-frequency models under different economic conditions, highlighting the importance of these models during periods of high economic uncertainty. Despite the usefulness of mixed-frequency models, there has been limited research on their application to house prices. Most studies on MIDAS and MF-VAR focus on forecasting GDP, for example, Clements and Galvão (2008, 2009), Marcellino and Schumacher (2010), Kuzin, Marcellino and Schumacher (2011), Schorfheide and Song (2015), Franta, Havrlant and Rusnák (2016) and Kuck and Schweikert (2021). MIDAS and its expansions have also been

explored for financial applications by Ghysels, Santa-Clara and Valkanov (2005, 2006), Ghysels, Rubia and Valkanov (2009), Le (2020) and Gong, Bu and Chen (2020). Another notable gap in the literature can be found in the literature on nowcasting focusing on the housing market. Stundziene et al. (2023) find that most of these studies (68.9%) focus on nowcasting GDP in their systematic review of 200 papers on nowcasting in the Web of Science (WoS) Core Collection and Scopus databases. Among these, only three papers address house prices: Kolbe et al. (2021) compute nowcasts by regressing sale price growth on lags of sale and ask prices. Sayag, Ben-Hur and Pfeffermann (2022) develop nowcasting models to address the effects of late-reported transactions on provisional HPIs. Van Veldhuizen, Vogt and Voogt (2016) forecast monthly housing transactions in the Netherlands using weekly Google searches of the word "mortgage" in a linear regression model estimated by Ordinary Least Squares (OLS). However, house prices are the leading economic indicators for overall economic health, directly impacting household wealth and living conditions. Therefore, this study extends the existing literature by applying mixed-frequency models to the housing market. Unlike the models in Kolbe et al. (2021) and Sayag, Ben-Hur and Pfeffermann (2022), MIDAS models and MF-VAR enable timely nowcasting of house prices by incorporating various predictors sampled at a high-frequency. Furthermore, by combining series of predictors, this study provides more comprehensive nowcasts and forecasts of house prices.

The paper proceeds as follows. Section 2 provides a brief literature review, organized according to the stages of development of MIDAS and MF-VAR models. Section 3 describes MIDAS and MF-VAR, discusses their relative advantages and details the statistical software used in the paper. Section 4 presents the dataset, application design and model specification, followed by a discussion of the empirical results. Finally, Section 5 concludes.

2 Literature Review

This section provides a brief literature review, structured according to the progression of MIDAS and MF-VAR models. These state-of-the-art mixed-frequency models are designed to manage the

complexities of unbalanced datasets that arise from different sampling frequencies and missing values due to publication lags.

Ghysels, Santa-Clara and Valkanov (2004) introduce Mixed Data Sampling (MIDAS) regression models to handle time series data sampled at different frequencies. These models address practical scenarios where high-frequency data is used to predict a lower-frequency variable, such as using daily market data to forecast monthly economic variables. MIDAS regressions are advantageous because they utilize more information without pre-filtering, unlike traditional methods that aggregate or interpolate data. The paper highlights the efficiency of MIDAS models compared to distributed lag models. MIDAS models have been widely applied to financial data to explore the relationship between the conditional mean and the conditional variance of stock market returns or future volatility. Ghysels, Santa-Clara and Valkanov (2005) investigate the intertemporal relationship between the conditional mean and variance of stock market returns using MIDAS. They find a positive and significant risk-return trade-off, demonstrating that MIDAS outperforms rolling window and GARCH estimators. The study highlights the flexibility of MIDAS in incorporating high-frequency data to forecast monthly variance. Ghysels, Santa-Clara and Valkanov (2006) explore various MIDAS regressions to predict volatility using return data sampled at different frequencies. They find that daily realized power is the best predictor of future volatility, outperforming models based on realized volatility. The use of higher-frequency data, such as five-minute intervals, does not improve predictions. The study also highlights that daily lags of one to two months are sufficient to capture volatility persistence. Ghysels, Rubia and Valkanov (2009) analyze various multi-period volatility forecasting techniques, including direct, iterated and MIDAS, using data from the US stock market. Their findings indicate that direct forecasting methods are less effective, iterated methods perform optimally for short-term horizons, and MIDAS is superior for long-term horizons, especially at very long horizons. Körs and Karan (2023) conduct a comparative analysis of the GARCH model and MIDAS with a beta polynomial weighting function against option-implied volatility (IV) indexes during the 2008 financial crisis. They assess the one-month out-of-sample volatility forecast performance of these models. Their findings indicate that MIDAS provides superior

forecasts compared to the GARCH model and IV method and that MIDAS is a sophisticated tool for forecasting future volatility due to its ability to process high-frequency data.

MIDAS models have also been used to forecast variables that reflect economic growth using macroeconomic predictors. In macroeconomic applications, Clements and Galvão (2008) introduce autoregressive dynamics as a common factor in MIDAS. Their real-time forecasting exercise shows that MIDAS, incorporating AR terms and within-quarter information from monthly predictors, can significantly reduce root mean square error (RMSE) compared to traditional quarterly-frequency AR or AR distributed lag models. Furthermore, Clements and Galvão (2009) assess the predictive power of leading indicators for predicting output growth over a one-year horizon using the MIDAS regression approach. They find that MIDAS, which combines multiple predictors in a parsimonious model, significantly enhances forecast accuracy, especially when using real-time vintage data. The study demonstrates that including monthly data available within the quarter improves predictions and that combining predictors within the MIDAS framework is more effective than forecast combination methods. Marcellino and Schumacher (2010) introduce the Factor MIDAS approach and apply it to nowcasting and forecasting German quarterly GDP using a large monthly dataset of the German economy. Their findings indicate that the Factor MIDAS models perform better than quarterly models, confirming that mixed-frequency models are useful for exploiting timely information from business cycle predictors.

Foroni, Marcellino and Schumacher (2015) argue that imposing restrictions on lag polynomials may be unnecessary, as parameter proliferation is not a concern when the difference in sampling frequencies is small. With this in mind, they developed the Unrestricted MIDAS (U-MIDAS) model within a linear framework. Through Monte Carlo simulations and empirical applications to out-of-sample nowcasting of GDP in the Euro area and the US using monthly predictors, they find that U-MIDAS generally outperforms MIDAS for forecasting quarterly dependent variables with monthly explanatory variables. However, when the difference in sampling frequencies is larger, MIDAS performs better than U-MIDAS.

Incorporating daily data allows for more frequent updates to economic forecasts, making it possible

to update forecasts when new information becomes available, which is crucial for real-time economic analysis. Andreou, Ghysels and Kourtellos (2013) demonstrate the use of MIDAS for forecasting quarterly economic activity with daily financial data. The model employs distributed lag polynomials to incorporate high-frequency data. Despite the overall decline in predictive power after the financial crisis, MIDAS exhibits smaller losses and enhances forecast accuracy. This study also highlights the critical role of high-frequency financial data in improving real-time economic forecasts.

Studies on MF-VAR models have been conducted in both frequentist and Bayesian contexts. Within a frequentist context, Mittnik and Zadrozny (2005) present and assess a Kalman filtering method for forecasting quarterly German real GDP on a monthly basis. They employ VAR(2) models that incorporate quarterly GDP and up to three monthly predictor variables. These models are estimated using a Kalman-filtering-based maximum likelihood method, in which the Kalman filter developed by Zadrozny is used to compute the likelihood in estimation and to generate monthly GDP forecasts. The results demonstrate that the monthly GDP forecasts for the first two months of a quarter are more accurate than one-quarter-ahead GDP forecasts obtained by the same model with quarterly data. Mariano and Murasawa (2010) compare mixed-frequency VAR and factor model for latent monthly real GDP and other monthly predictors. They set up a VAR model for partially latent monthly series and derive a state-space model, in which the mixed-frequency series are treated as monthly series with missing observations. Maximum likelihood estimation is employed for a linear Gaussian state-space model with missing observations. The empirical findings indicate that the relative performance of MF-VAR and the factor model depends on the underlying data generation process.

Eraker et al. (2014) propose a Bayesian estimation of MF-VAR models. To draw a posterior distribution, they develop a Gibbs sampler that produces draws of latent variables and parameters. In their simulation work, they show that taking into account mixed-frequency data leads to smaller RMSEs for all parameter estimates. Schorfheide and Song (2015) develop MF-VAR estimated with Bayesian methods with a Minnesota prior to cope with the high dimensionality of the parameter space. Using a real-time dataset with three quarterly and eight monthly variables, they evaluate

the forecasting performance of MF-VAR compared to those of a standard quarterly frequency VAR. Their findings indicate that incorporating monthly variables significantly enhances short-run forecast accuracy, resulting in considerably lower RMSE. One of the Bayesian developments of VAR is the steady-state prior proposed by Villani (2009). The steady-state prior is based on a mean-adjusted form of the VAR where the unconditional mean of the series is explicitly parameterized. Ankargren, Unosson and Yang (2018) present a mixed-frequency VAR with a steady-state (the unconditional mean) prior. Using a real-time dataset, they forecast Swedish GDP growth and find that the MF-VAR with a steady-state prior improves forecasting accuracy compared to Bayesian quarterly and mixed-frequency VARs without the steady-state prior. Additionally, they also find that the high-frequency data improves the accuracy of low-frequency forecasts, particularly for shorter time horizons.

Kuzin, Marcellino and Schumacher (2011) compare the nowcasting performance of MIDAS models (both with and without AR terms) and the MF-VAR model presented by Mariano and Murasawa (2010) for GDP growth and monthly predictors. They carry out nowcasting of monthly GDP growth using the Kalman smoother and evaluate the performance of mixed-frequency models by calculating the relative MSFEs, derived from a recursive estimation of an AR model for monthly forecast horizons ranging from one to nine months. Their empirical results indicate that MF-VAR outperforms for longer nowcast horizons, while MIDAS performs better for shorter horizons. Foroni and Marcellino (2014) evaluate the performance of bridge equations, AR-MIDAS and MF-VAR in nowcasting the quarterly growth rate of Euro area GDP, utilizing a comprehensive set of monthly indicators (approximately 150 series). They use the Kalman filter to derive monthly estimates of the quarterly series. Their analysis is conducted recursively within a pseudo-real-time framework, assessing the nowcasting performance of the models by MSFEs. The results indicate that MIDAS with an AR component performs notably well, often outperforming MF-VAR across most horizons. Bridge equations also demonstrate strong overall performance.

3 Methodology

In this section, I describe the mixed-frequency models used in this study: the mixed-data sampling (MIDAS) and mixed-frequency vector autoregression (MF-VAR) approaches. The set of MIDAS models includes the basic MIDAS, its expansion with an AR component and its unrestricted variant (U-MIDAS). MF-VAR incorporates Bayesian methods. Furthermore, I discuss their relative advantages and introduce the evaluation metrics and statistical software used for their empirical application in this paper.

In terms of notation, t represents the basic time unit at the high-frequency (daily or monthly) and m represents a ratio between the high and low sampling frequency. For example, for quarterly HPI growth and a monthly predictor, m is equal to three. The quarterly German HPI growth is written as $y_{q,t}$ for $t = \{m, 2m, 3m, \cdots, T_q\}$, where T_q is the final quarter for which the HPI growth is available. A predictor sampled at the higher-frequency is denoted as $y_{m,t}$ for $t = \{1, 2, 3, \cdots, T_m\}$, where T_m is the final period (day or month) for which the predictor is available. Since the predictor is available earlier than HPI observations, T_m is generally larger than T_q. Given the available information set, the forecast for HPI is denoted as $\hat{y}_{q,T_q} = y_{q,T_q+h|T_m}$.

3.1 Mixed-data sampling

To cope with mixed frequency data, Ghysels, Santa-Clara and Valkanov (2004) propose MIDAS, in which the dependent variable $y_{q,t}$ sampled at the low-frequency is regressed on a distributed lag of $y_{m,t}$, which is sampled at the higher-frequency. The MIDAS regression is tightly parameterized with highly parsimonious distributed lag polynomials to prevent the proliferation of parameters. Clements and Galvão (2008) expanded MIDAS by introducing autoregressive (AR) dynamics as a common factor. Introducing autoregressive dynamics into MIDAS models is desirable, as AR models often produce competitive forecasts compared to those of models that include explanatory variables. Foroni, Marcellino and Schumacher (2015) derived unrestricted MIDAS (U-MIDAS) regressions within a general linear dynamic framework, without restriction on the lag polynomials.

The parameters can be easily estimated using ordinary least squares (OLS). This computational simplicity is U-MIDAS's greatest advantage.

3.1.1 The basic MIDAS

For forecast horizon h, the basic MIDAS for a single explanatory variable is given by

$$y_{q,t+h} = \beta_0 + \beta_1 B(L;\theta) y_{m,t+w} + \varepsilon_{q,t+h}, \tag{1}$$

where lag polynomial $B(L;\theta) = \sum_{k=0}^{K} b(k;\theta) L^k$ with the lag operator at the higher-frequency such that $Ly_{m,t} = y_{m,t-1}$. $y_{m,t}$ is the skip-sampled predictor sampled at the higher-frequency with $w = T_m - T_q$.

As the regressors $y_{m,t}$ are observed at a higher-frequency than $y_{q,t}$, overfitting can easily occur in the unrestricted linear case. Ghysels, Sinko and Valkanov (2007) discuss several non-linear weighting schemes for a parsimonious parametrization of the lagged coefficient $b(k;\theta)$. In this paper, I employ an exponential Almon lag defined as

$$b(k;\theta) = \frac{exp(\theta_1 k + \cdots + \theta_Q k^Q)}{\sum_{k=1}^{K} exp(\theta_1 k + \cdots + \theta_Q k^Q)}.$$

This function form is quite flexible, allowing for parameters to take various shapes such as decreasing, increasing, or hump-shaped patterns with only a few parameters. In line with Clements and Galvão (2009), the number of parameters to be estimated is set to two, that is, $Q = 2$. The quadratic exponential lag function $B(L;\theta)$ is not only parsimonious but also allows for large lag length K in approximating the impulse response function of the low-frequency variable to the predictor at the higher-frequency.

MIDAS can be estimated using nonlinear least squares (NLS) with lag order K specified by the Bayesian information criterion (BIC), yielding the coefficients $\widehat{\theta}_1$, $\widehat{\theta}_2$, $\widehat{\beta}_0$ and $\widehat{\beta}_1$. Thus, the forecast is given by

$$y_{q,T_q+h|T_x} = \widehat{\beta}_0 + \widehat{\beta}_1 B(L;\widehat{\theta}) y_{m,T_m},$$

where the last observation y_{m,T_m} contains the most recent values of $y_{m,t}$ for $T_m = T_q + w$. Since MIDAS is h-dependent, re-estimation is required for multi-step forecasts. This is also necessary whenever new data, such as a new monthly observation within a quarter, becomes available.

3.1.2 The Autoregressive MIDAS

As pointed out by Ghysels, Santa-Clara and Valkanov (2005), introducing autoregressive dynamics into MIDAS models is not straightforward. Directly adding AR terms to the basic MIDAS model results in efficiency losses. Moreover, this can induce a seasonal response in the dependent variable to the explanatory variable, regardless of whether the dependent variable exhibits seasonal patterns (Clements and Galvão, 2008). Therefore, Clements and Galvão (2008) introduce an extension to the basic MIDAS approach, in which the AR coefficients and the lagged coefficients are restricted by a common-factor restriction. This extension, denoted as AR-MIDAS, is defined as

$$y_{q,t+h} = \beta_0 + \tau y_{q,t} + \beta_1 B(L;\theta)(1 - \tau L^m) y_{m,t+w} + \varepsilon_{q,t+h}. \tag{2}$$

In this framework, the response of $y_{q,t}$ to $y_{m,t}$ remains non-seasonal. The AR dynamics coefficients τ can be estimated by NLS along with other coefficients. Given estimated parameters $\widehat{\theta}_1$, $\widehat{\theta}_2$, $\widehat{\beta}_0$, $\widehat{\beta}_1$, $\widehat{\tau}$, the AR-MIDAS forecast is computed as

$$y_{q,T_q+h|T_m} = \widehat{\beta}_0 + \widehat{\tau} y_{q,T_q} + \widehat{\beta}_1 B(L;\widehat{\theta})(1 - \widehat{\tau} L^h) y_{m,T_m}.$$

Regarding the weighting scheme for parameterization in AR-MIDAS, I retain the quadratic exponential Almon lag polynomial, as used in the basic MIDAS.

3.1.3 The Unrestricted MIDAS

Foroni, Marcellino and Schumacher (2015) developed the U-MIDAS model within a general linear dynamic framework, without imposing any functional form restrictions on the lag polynomials. Furthermore, the AR terms are not also restricted by the common factor as in Clements and Galvão

(2008). The model is therefore expressed as

$$y_{q,t+h} = \alpha(L)y_{q,t} + \delta(L)y_{m,t+w} + \varepsilon_{q,t+h}, \tag{3}$$

where $\alpha(L) = \sum_{p=1}^{P} \alpha_p L^{mp}$ and $\delta(L) = \sum_{k=0}^{K} \delta_k L^k$. The parameters can be estimated by OLS, with the lag orders P and K specified by the BIC. In their Monte Carlo experiments, Foroni, Marcellino and Schumacher (2015) find that forecasting models selected by the AIC perform worse than those chosen by the BIC. The AIC generally selects a larger lag order than the BIC, increasing the number of parameters to be estimated and potentially leading to overparameterization.

Given the coefficients $\widehat{\alpha}_p$ and $\widehat{\delta}_k$ for lag length P and K, respectively, the U-MIDAS forecast is computed as

$$y_{q,T_q+h|T_m} = \widehat{\alpha}(L)y_{q,T_q} + \widehat{\delta}(L)y_{m,T_m}.$$

3.2 Mixed-frequency vector autoregression

MF-VAR is a system approach assumed to evolve at the highest available frequency, which is monthly in this paper, resulting in missing monthly observations for the quarterly German HPI. To cope with these missing observations, the MF-VAR is represented as a state-space model. The state equation describes the underlying, potentially unobserved, monthly series of the quarterly variables as the latent states of the system. The measurement equation relates the observed series to the underlying monthly variables. In this study, I equip the MF-VAR with a Bayesian framework. The prior information shrinks the model parameters, reducing the risk of overfitting. Additionally, I employ the steady-state prior introduced by Villani (2009). Most priors used in VAR models focus primarily on the dynamic coefficients while providing limited information about the deterministic component of the model. However, incorporating an informative prior for the unconditional mean, also known as the steady state, is crucial, as stationary VARs converge to this mean over long horizons. Including this prior information can potentially improve the accuracy of the model. The application of a steady-state prior in MF-VAR is based on the approach used by Ankargren, Unosson

and Yang (2018).

3.2.1 State-space representation of MF-VAR

Consider the system that evolves at the monthly frequency. Define $z_t = (z'_{m,t}, z'_{q,t})'$ as the underlying monthly vector in the system, consisting of n_m monthly $z_{m,t}$ and n_q the unobserved latent variables quarterly variable $z_{q,t}$, with a dimension $n = n_m + n_q$. Note that the time t takes the monthly frequency. Additionally, let y_t represent the observations at t, with a dimension n_t. The empirical challenge is that the observed data changes over time, that is, n_t is not always equal to n.

The underlying monthly VAR(r) for z_t is given by

$$\Pi(L)z_t = \Phi c_t + u_t, \quad u_t \sim N_n(0, \Sigma), \tag{4}$$

where $\Pi(L) = (I_n - \Pi_1 L - \cdots - \Pi_r L^r)$ is a r-th order invertible lag polynomial, c_t is a $d \times 1$ vector of deterministic components and Φ is an $n \times d$ matrix of parameters.

The model in (4) refers to the standard VAR. Following the approach of Villani (2009), this can be represented in mean-adjusted form as

$$\Pi(L)(z_t - \Psi c_t) = u_t, \quad u_t \sim N_n(0, \Sigma), \tag{5}$$

where $\Psi = [\Pi(L)]^{-1}\Phi$, if the process is stationary. By the reformulation, the unconditional mean of z_t is directly specified by Ψ as $E(z_t) = \Psi c_t := \mu_t$.

By defining $Z_t = (z'_t, \ldots, z'_{t-r+1})$, the Equation (5) can be written in companion form as

$$Z_{t+1} = W_{t+1}\psi + F(\Pi)(Z_t - W_t\psi) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \Omega(\Sigma)), \tag{6}$$

where

$$W_t = [(c_t \otimes I_r), \ldots, (d_{t-r+1} \otimes I_r)]', \quad \psi = \text{vec}(\Psi), \quad \Pi = (\Pi_1, \ldots, \Pi_r)$$

$$F(\Pi) = \begin{pmatrix} \Pi \\ I_{n(r-1)} \quad 0_{n(r-1)\times r} \end{pmatrix}, \qquad \Omega(\Sigma) = \begin{pmatrix} \Sigma & 0_{n\times n(r-1)} \\ 0_{n(r-1)\times n} & 0_{n(r-1)} \end{pmatrix}.$$

Equation (6) constitutes the state equation of MF-VAR.

To describe the measurement equation, let T denote the forecast origin and let $T_q \leq T$ indicate the last period corresponding to the final month of the quarter. The monthly series is observed every month up to period T_q. Hence, for $t = \{1, \ldots, T_q\}$, the observations of monthly variable is denoted as

$$y_{m,t} = z_{m,t}. \tag{7}$$

Furthermore, let Λ_q and M_t be deterministic aggregation and selection matrices, respectively. The aggregation matrix Λ_q is based on the selected weighting scheme for aggregating the unobserved monthly latent observations in $z_{q,t}$ into the low-frequency observation $y_{q,t}$. In this study, I employ the inter-quarter average as the aggregation scheme,[1] such that if the underlying monthly VAR has at least three lags,

$$\tilde{y}_{q,t} = \frac{1}{3}(z_{q,t} + z_{q,t-1} + z_{q,t-2}) = \Lambda_q Z_t.$$

Note that $\tilde{y}_{q,t}$ is observed only every three month. Let $M_{q,t}$ be identity matrix I_{n_q} if t corresponds to the last month of a quarter such that $y_{q,t} = \tilde{y}_{q,t}$. Otherwise, $M_{q,t}$ is an empty matrix such that $y_t = y_{m,t}$. Therefore, for $t = \{1, \ldots, T_q\}$,

$$y_{q,t} = M_{q,t}\tilde{y}_{q,t} = M_{q,t}\Lambda_q Z_t. \tag{8}$$

For periods $t = \{T_q + 1, \ldots, T\}$, no further observations of the quarterly variable are obtained. However, some monthly variables can be observed. Let $M_{m,t}$ be a deterministic selection matrix such that, for $t = \{T_q + 1, \ldots, T\}$,

$$y_{m,t} = M_{m,t}\Lambda z_{m,t}. \tag{9}$$

[1] Other aggregation schemes are also possible; for example, see Mariano and Murasawa (2010).

Note that the availability of monthly observations varies over time. Finally, Equations (7), (8) and (9) establish the measurement equation such that, for $t = \{1, \ldots, T\}$,

$$y_t = M_t \Lambda Z_t, \quad t = \{1, \ldots, T\}, \tag{10}$$

where M_t is a sequence of selection matrices. Adjusting M_t reflects the time-varying availability of monthly observations over time by selecting variables observed by period T. Specifically, if the monthly variable $z_{m,t}$ is missing at time t, the corresponding row in M_t is dropped.

3.2.2 Prior beliefs

In Bayesian inference, it is essential to specify prior distributions for the model parameters. In this paper, the parameters of interest are Π, Σ and Ψ in Equations (5) and (6). Following Villani (2009) and Ankargren, Unosson and Yang (2018), I consider a normal prior for the parameters in Ψ and a normal-inverse Wishart as a joint prior for Π and Σ. Thus, the prior used is

$$(\Pi, \Sigma) \sim MNIW(\underline{\Pi}, \underline{\Omega}_\Pi, \underline{S}, \underline{\nu}),$$

such that

$$\Sigma \sim IW(\underline{S}, \underline{\nu}), \quad \text{vec}(\Pi') | \Sigma \sim N_{n^2 r}(\text{vec}(\underline{\Pi'}), \Sigma \otimes \underline{\Omega}_\Pi).$$

The structure of the moments of the prior distribution is specified along the lines of the Minnesota prior beliefs, resulting in

$$\begin{aligned} \underline{\Pi}(\gamma) &= \begin{pmatrix} \text{diag}(\underline{\gamma}) & 0_{n \times [(r-1)+1]} \end{pmatrix}' \\ \underline{\omega}_{ii} &= \frac{\lambda_1^2}{(l^{\lambda_2} s_a)^2} \quad \text{for lag } l \text{ of variable } a,\ i = (l-1)r + a, \end{aligned}$$

where $\underline{\omega}_{ii}$ are the diagonal elements of $\underline{\Omega}_\Pi$ and the prior mean of AR(1) parameters $(\underline{\gamma})$ is set to 0.9, while the prior means of the other regression parameters are set to zero. λ_1 is the overall tightness and λ_2 determines the lag decay rate. The inclusion of s_j for $j = \{1, \ldots, r\}$ is for adjusting

differences in the measurement scale of the variables. For n variables, the prior parameter matrix of the inverse Wishart is

$$\underline{S} = (\underline{\nu} - n - 1)\ diag\ (s_1^2, \ldots, s_n^2),$$

where s_i^2 are n residual variances from auxiliary AR(4) regressions. The degree of freedom $\underline{\nu}$ is fixed to $n+2$ to ensure that the prior variance exists (Kadiyala and Karlsson, 1997).

Finally, following Villani (2009), the prior for the unconditional mean is given by

$$\psi = \text{vec}(\Psi) \sim N_{nm}(\underline{\psi}, \underline{\Omega}_{\psi}).$$

The steady-state priors are defined by 95% probability intervals for the variables.

3.2.3 Posterior inference

A Gibbs sampler is employed to draw from the intractable posterior distribution of latent variables and parameters given the data, $p(\Pi, \Sigma, \psi, Z|Y)$. This approach decomposes the posterior into three blocks $p(Z|\Pi, \Sigma, \psi, Y)$, $p(\Pi, \Sigma|\psi, Z)$ and $p(\psi|\Pi, \Sigma, \psi, Z)$ of full conditional densities, making the sampling process straightforward. It can be observed that the parameters (Π, Σ) and ψ are independent of Y given Z.

Algorithm 1 presents the main steps of the Gibbs sampler for $e = \{1, \ldots, Sim\}$, where *Sim* denotes the number of simulations. Initially, a simulation smoother is employed to sample the latent data conditional on the parameters, as detailed by Durbin and Koopman (2002, 2012). The subsequent steps involve drawing samples from the normal-inverse-Wishart and multivariate normal distributions, respectively.

Algorithm 1 Gibbs sampler for mixed-frequency steady-state Bayesian VAR

1: **for** $e = 1$ to *Sim* **do**
2: Draw $Z^{(e)}$ from $p(Z|\Sigma^{(e-1)}, \Pi^{(e-1)}, \Psi^{(e-1)})$
3: Draw $(\Pi, \Sigma)^{(e)}$ from $p(\Pi, \Sigma|\Psi^{(e-1)}, Z^{(e)})$
4: Draw $\psi^{(e)}$ from $p(\psi|\Pi^{(e)}, \Sigma^{(e)}, Z^{(e)})$
5: **end for**

To draw the unobserved variables, let W_t be as in the state equation (6) so that $W_t\psi = E(Z_t)$ and $Z_t^* = (Z_t - W_t\psi)$. Considering the measurement equation (10), it can be observed that

$$y_t^* \equiv y_t - M_t\Lambda W_t\psi = M_t\Lambda(Z_t - W_t\psi) \equiv M_t\Lambda Z_t^*$$

is the same as $y_t = M_t\Lambda Z_t$. Hence, the state-space model is equivalent to

$$\begin{aligned} y_t^* &= M_t\Lambda Z_t^* \\ Z_{t+1}^* &= F(\Pi)Z_t^* + \varepsilon_{t+1}, \quad \varepsilon_t \sim N(0, \Omega(\Sigma)). \end{aligned} \tag{11}$$

Thus, it is sufficient to provide the Kalman filter without deterministic components in the following discussion, as the necessary Kalman filter steps can be performed using y_t^* and Z_t^*, with the appropriate deterministic term added afterward.

According to the simulation smoother by Durbin and Koopman (2002), assume the state-space system in (11) for the period $t = 1, \cdots, T$ and treat the parameters as fixed and known. Applying the Kalman filter means to apply the following equations recursively

$$\begin{gathered} O_t = F(\Pi)O_{t-1} + K_{t-1}\upsilon_{t-1}, \quad P_t = F(\Pi)P_{t-1}L_{t-1}' + \Omega(\Sigma), \quad \upsilon_t = y_t - M_t\Lambda O_t, \\ F_t = M_t\Lambda P_t\Lambda' M_t', \quad K_t = F(\Pi)P_t\Lambda' M_t' F_t^{-1}, \quad L_t = F(\Pi)K_t M_t\Lambda \end{gathered} \tag{12}$$

for $t = 1, \cdots, T$. Note that the dimensions of υ_t, F_t and K_t vary deterministically as a function of observations available at time t. The smoothed state vector is computed by first evaluating

$$R_{t-1} = M_t\Lambda F_t^{-1}\upsilon_t \tag{13}$$

backwards, that is, for $t = T, T-1, \cdots, 1$ with $R_T = 0$. Then, the smoothed mean of the latent state $\hat{Z}_t = E(Z_t|Y_{1:T})$ is computed by applying the forwards recursion

$$\hat{Z}_{t+1} = F(\Pi)\hat{Z}_t + \Omega(\Sigma)R_t, \tag{14}$$

which is initialized by $\hat{Z}_1 = a_1 + P_1 R_0$. To draw from the density $p(Z_{1:T}|Y_{1:T})$, pseudo-variables y^+ and Z^+ are generated by drawing ε_t from $N(0,\Omega(\Sigma))$ and using the recursion in (11). Given these pseudo-variables, the latent state is smoothed, which yield $\hat{Z}^+ = E(Z^+_{1:T}|Y^+_{1:T})$. The final draw is then obtained by computing

$$\check{Z}_{1:T} = \hat{Z}_{1:T} + \left(Z^+_{1:T} - \hat{Z}^+_{1:T}\right).$$

Note that $\hat{Z}_{1:T}$ is obtained by first applying the Kalman smoother based on y^*_t and Z^*_t, which yields $\hat{Z}^*_{1:T}$. Then, $\hat{Z}_{1:T} = \hat{Z}^*_{1:T} + (I_N \otimes \psi')W_{1:T}$, where $W_{1:T} = (W_1, \cdots, W_T)'$.

Algorithm 2 summarizes the steps of the simulation smoother. When conditioning on latent states

Algorithm 2 Simulation smoother

1: Draw $\varepsilon^+_1, \cdots, \varepsilon^+_T$ from $N(0,\Omega(\Sigma))$ and use (11) to construct $Y^+_{1:T}$ and $Z^+_{1:T}$
2: Compute $\hat{Z}^*_{1:T} = E(Z^*_{1:T}|Y^*_{1:T})$ and $\hat{Z}^+_{1:T} = E(Z^+_{1:T}|Y^+_{1:T})$ by applying (12) forwards, (13) backwords and (14)
3: Compute the final draw as $\check{Z}_{1:T} = \hat{Z}^*_{1:T} + (I_N \otimes \psi')W_{1:T} + \left(Z^+_{1:T} - \hat{Z}^+_{1:T}\right)$

and unconditional mean, the model is a standard VAR (4) for $(z_t - \Psi c_t)$ and the resulting posteriors follow standard results (see Karlsson, 2013 for more details). Hence, the posterior distribution for the dynamic coefficients is

$$\text{vec}(\Pi')|Z,\Sigma,\psi \sim N(\text{vec}(\overline{\Pi}'), \Sigma \otimes \overline{\Omega}_\Pi),$$

or, equivalently

$$\Pi'|Z,\Sigma,\psi \sim MN(\overline{\Pi}', \Sigma \otimes \overline{\Omega}_\Pi),$$

where $\overline{\Omega}^{-1}_\Pi = \underline{\Omega}^{-1}_\Pi + \check{Z}'_{1:T-1}\check{Z}_{1:T-1}$ and $\overline{\Pi} = \overline{\Omega}_\Pi(\underline{\Omega}^{-1}_\Pi \underline{\Pi} + \check{Z}'_{1:T-1}\check{z}_{2:T})$. The demeaned z_t in non-companion form can be written as

$$\check{z} = \begin{pmatrix} z'_1 - d'_1\Psi' \\ \vdots \\ z'_T - d'_T\Psi' \end{pmatrix} = \begin{pmatrix} z'_1 \\ \vdots \\ z'_T \end{pmatrix} - \begin{pmatrix} d'_1\Psi' \\ \vdots \\ d'_T\Psi' \end{pmatrix} = z - \begin{pmatrix} \psi'(d_1 \otimes I_r) \\ \vdots \\ \psi'(c_t \otimes I_r) \end{pmatrix}.$$

Thus, we have the companion form as follows

$$\check{Z}_t = \begin{pmatrix} \check{z}_t \\ \check{z}_{t-1} \\ \vdots \\ \check{z}_{t-r+1} \end{pmatrix}, \quad \check{Z}_{1:T} = \begin{pmatrix} \check{Z}_1' \\ \check{Z}_2' \\ \vdots \\ \check{Z}_T' \end{pmatrix} = \begin{pmatrix} \check{z}_1 & \check{z}_0 & \cdots & \check{z}_{2-r} \\ \check{z}_2 & \check{z}_1 & \cdots & \check{z}_{3-r} \\ \vdots & \vdots & \ddots & \vdots \\ \check{z}_T & \check{z}_{T-1} & \cdots & \check{z}_{T-r+1} \end{pmatrix}.$$

The posterior for the error covariance is

$$\Sigma | Z, \psi \sim IW(\overline{S}, \overline{\nu}),$$

where

$$\begin{aligned}
\overline{\nu} &= T + \underline{\nu}, \\
\overline{S} &= \underline{S} + S + (\underline{\Pi} + \widehat{\Pi})' \left(\underline{\Omega}_{\Pi} + (\check{Z}_{1:T-1}' \check{Z}_{1:T-1})^{-1} \right)^{-1} (\underline{\Pi} + \widehat{\Pi}), \\
\widehat{\Pi} &= (\check{Z}_{1:T-1}' \check{Z}_{1:T-1})^{-1} (\check{Z}_{1:T-1} \check{z}_{2:T}), \\
S &= (\check{z}_{2:T} - \check{Z}_{1:T-1} \widehat{\Pi})' (\check{z}_{2:T} - \check{Z}_{1:T-1} \widehat{\Pi}).
\end{aligned}$$

Finally, to derive the conditional posterior of Ψ, let

$$\check{Y} = \begin{pmatrix} z_1'\Pi(L)' \\ \vdots \\ z_T'\Pi(L)' \end{pmatrix} = \begin{pmatrix} z_1' - z_0'\Pi_1' - z_{-1}'\Pi_2' - \cdots - z_{-r+1}'\Pi_r' \\ \vdots \\ z_T' - z_{T-1}'\Pi_1' - z_{T-2}'\Pi_2' - \cdots - z_{T-r}'\Pi_r' \end{pmatrix} = \begin{pmatrix} z_1' - \begin{pmatrix} z_0' & \cdots & z_{-r+1}' \end{pmatrix}\Pi \\ \vdots \\ z_T' - \begin{pmatrix} z_{T-1}' & \cdots & z_{T-r}' \end{pmatrix}\Pi \end{pmatrix}$$

$$= \begin{pmatrix} z_1' - Z_0'\Pi \\ \vdots \\ z_T' - Z_{T-1}'\Pi \end{pmatrix} = z_{1:T} - Z_{0:T-1}\Pi,$$

$$D = \begin{pmatrix} D_1^- \\ \vdots \\ D_T^- \end{pmatrix}, \quad D_t^- = \begin{pmatrix} d_t' & -d_{t-1}' & \cdots & -d_{t-r}' \end{pmatrix}, \quad \Theta = \begin{pmatrix} \psi & \Pi_1\psi & \cdots & \Pi_r\psi \end{pmatrix}.$$

The mean-adjusted model (5) can be written as

$$\Pi(L)z_t = \Pi(L)\Psi c_t + u_t = \Psi d_t - \Pi_1\Psi d_{t-1} - \cdots - \Pi_r\Psi d_{t-r} + u_t$$

such that

$$\check{Y} = D\Theta + U,$$

where $U = (u_1, \cdots, u_T)'$. The conditional posterior $\psi|Z,\Sigma,\Pi$ is derived from multivariate regression and hence

$$\text{vec}(\Theta') = E\psi = \begin{pmatrix} I_{rm} \\ I_m \otimes \Pi_1 \\ \vdots \\ I_m \otimes \Pi_r \end{pmatrix} \psi,$$

$$\psi|Z,\Pi,\Sigma \sim N(\overline{\psi}, \overline{\Omega}_\psi),$$

where $\overline{\Omega}_\psi = E'(D'D \otimes \Sigma^{-1})E + \underline{\Omega}_\psi^{-1}$ and $\overline{\psi} = \overline{\Omega}_\psi(E'\text{vec}(\Sigma^{-1}Y'D) + \underline{\Omega}_\psi^{-1}\underline{\psi})$.

3.2.4 Hyperparameter selection

Selecting hyperparameters in a Bayesian VAR (BVAR) model is critical to ensure its performance and accuracy. This study follows the approach suggested by Robertson and Tallman (1999), which involves evaluating the forecast performance of the model across a range of hyperparameter values. This method helps in identifying a model that avoids overfitting. However, it is essential to determine the appropriate forecast horizon and the number of forecasts to be used during the hyperparameter selection process.

By varying the hyperparameters and expanding the estimation sample recursively, MF-VAR can better capture the dynamics of economic variables. This study allows hyperparameters to vary over time as the estimation sample expands. The selection process is based on the accuracy of the nowcasts. For each forecast origin, I nowcast the German HPI with MF-VAR using various pairs of hyperparameters from the sets $\lambda_1 = \{0.01, 0.1, 0.5, 0.9\}$ and $\lambda_2 = \{0.1, 0.5, 1, 2, 4\}$. This process utilizes all available information in the evaluation sample. For instance, if the forecast origin is the 44th quarter and the selection is based on eight-step-ahead forecasting performance, the forecasting performance can be evaluated by comparing the differences between the forecasts and the observed values from the 52nd quarter onwards. Conversely, if the horizon is set to zero, all forecasts can be compared to all corresponding observations throughout the evaluation sample period.

3.3 A comparison of MIDAS and MF-VAR

Mixed-frequency models are developed to handle data sampled at different frequencies and incorporate timely observations available at higher-frequencies. However, there are distinct differences between these models. MIDAS, AR-MIDAS and U-MIDAS are single-equation approaches that allow for different frequencies between the dependent and explanatory variables. In contrast, MF-VAR is a system approach that jointly specifies the dynamics of variables. In MF-VAR, the system is assumed to evolve at the highest available frequency, leading to missing values for the higher-frequency observations of the low-frequency variables. To manage these missing values due to different frequencies, MF-VAR is represented as a state-space model.

MIDAS and AR-MIDAS are especially advantageous for analyzing high-frequency financial data. The application of restricted distributed lag polynomials facilitates the straightforward modeling of daily time series. For example, the quadratic exponential lag function is parsimonious and allows for the lag length to be large in approximating the impulse response function of the low-frequency variable to the predictor with the higher-frequency. Adding a daily variable to the explanatory variables requires only three additional coefficients (θ_1, θ_2 and β) in the lag polynomial. Conversely, modeling daily data in MF-VAR significantly increases the model's computational complexity due to the large number of parameters to be estimated.

While MIDAS and AR-MIDAS employ the non-linear weighting scheme for the lag coefficient, U-MIDAS is a linear regression model that does not impose restrictions on lag polynomials, providing substantial computational simplicity with parameter estimation by Ordinary Least Squares (OLS). Foroni, Marcellino and Schumacher (2015) demonstrated in their Monte Carlo experiments that U-MIDAS outperforms AR-MIDAS when the difference in sampling frequencies between variables is small. However, when U-MIDAS is applied to datasets with large differences in sampling frequency, it becomes heavily parameterized, resulting in poorer performance compared to AR-MIDAS. On the other hand, MF-VAR combined with Bayesian methods can use prior information to reduce the model parameters, helping to prevent overfitting.

Fundamentally, MIDAS, AR-MIDAS and U-MIDAS are frequentist approaches, while the MF-VAR model used in this study employs Bayesian methods. Both frequentist and Bayesian statistics aim to perform inference on a parameter vector, but their approaches differ. In the frequentist approach, unknown parameters are assumed to be fixed and available data are used to estimate the true vector. Conversely, Bayesian theory considers the parameter vector as a random variable associated with a probability distribution, called the posterior distribution, incorporating prior information on the parameter vector into the estimation process. To sample parameters and latent variables given the observed data, Markov Chain Monte Carlo (MCMC) methods, such as the Gibbs sampler, are used. In contrast, in MIDAS, it is only possible to obtain a high-frequency update of the expected low-frequency realization, not an estimate of the missing values in the low-frequency variable.

This discussion suggests that the advantages of these mixed-frequency models depend on the context of the application. Thus, their relative performance has to be evaluated through empirical applications. Chapter 4 of this paper focuses on the specific case of nowcasting and forecasting quarterly German HPI growth.

3.4 Forecast comparison

The accuracy of the forecasts yielded by the mixed-frequency models is assessed by comparing them to the actual observations within the evaluation sample. Let $t_e = \{1, \ldots, T_e\}$ denote the time index within the evaluation sample, where T_e represents the final quarter of this period. The forecasting error is defined as $\hat{\varepsilon}_{q,t_e} = y_{q,t_e} - \hat{y}_{q,t_e}$, where y_{q,t_e} are the actual observations and $\hat{y}_{q,t_e}$ are the forecasts. Forecast accuracy is evaluated using the mean squared forecasting error (MSFE) relative to a benchmark. When the differences in relative MSFEs are small, I employ the Diebold-Mariano (DM) test, as proposed by Diebold and Mariano (1995), to assess whether the differences in forecasting accuracy are statistically significant.

3.4.1 Mean squared forecasting error

The Mean Squared Forecasting Error (MSFE) and relative MSFE for each mixed-frequency model are computed as

$$\begin{aligned} \text{MSFE}_h &= \frac{1}{T_e} \sum_{t_e=1+h}^{T_e} \hat{\varepsilon}_{q,t_e}^2, \\ \text{Relative MSFE}_h &= \frac{\text{MSFE}_h(\text{model})}{\text{MSFE}_h(\text{benchmark})}. \end{aligned} \tag{15}$$

A relative MSFE greater than 1 indicates lower accuracy of the mixed-frequency model compared to the benchmark, whereas a value less than 1 indicates higher accuracy.

3.4.2 Diebold-Mariano test

Suppose that a pair of h-steps ahead forecasts errors $(\hat{\varepsilon}^A_{q,t}, \hat{\varepsilon}^B_{q,t})$ for $t_e = \{1+h, \ldots, T_e\}$. With a squared error function[2] $f(x) = x^2$, the loss differential series is defined as

$$\Delta_t = f(\hat{\varepsilon}^A_{q,t_e}) - f(\hat{\varepsilon}^B_{q,t_e}), \quad t_e = \{1+h, \ldots, T_e\}.$$

Δ_t is assumed to be a stationary time series. The null hypothesis of equal forecast accuracy can be represented as

$$H_0 : E[\Delta_t] = 0.$$

Denote the sample mean of Δ_t as $\bar{\Delta} = \frac{1}{T_e}\sum_{t_e=1}^{T_e}$, then the DM test statistic can be $\sqrt{T_e}\frac{\bar{\Delta}}{\hat{\sigma}_\Delta}$, which is asymptotically normal, that is,

$$\sqrt{T_e}\frac{\bar{\Delta}}{\hat{\sigma}_\Delta} \xrightarrow{d} N(0,1), \quad as \quad T_e \to \infty,$$

where $\hat{\sigma}_\Delta$ is an estimator of the standard deviation of Δ_t.

The DM test tends to reject the null hypothesis too often for small samples. To address this, Harvey, Leybourne and Newbold (1997) proposed a modified test, called the HLN test, where h-step ahead forecast errors are assumed to have zero autocorrelation at order h and beyond. The modified statistic is defined as

$$HLN = \left[\frac{T_e + 1 - 2h + h(h-1)}{T_e}\right]^{\frac{1}{2}} DM \tag{16}$$

3.5 Statistical software

The programming language `R` (version 4.3.3) is employed throughout the application process. The package `RJDemetra` (version 0.2.6) acts as the `R` interface to JDemetra+, the seasonal adjustment software officially recommended by the European Statistical System (ESS) and the European System of Central Banks. This package is utilized for the seasonal adjustment of time series

[2]Other loss functions can also be used, such as absolute errors.

data. The package `midasr` (version 0.8) offers methods and tools for analyzing mixed frequency time series data, allowing for estimation, model selection and forecasting for MIDAS regressions. The package `mfbvar` (version 0.5.6) provides functions and tools for estimating mixed-frequency Bayesian vector autoregressive (VAR) models, implementing a state space-based VAR model to cope with mixed data frequencies. MF-VAR is estimated using Markov Chain Monte Carlo (MCMC) methods to approximate the posterior distribution. The package `zoo` (version 1.8.12) is used for interpolating the national disposable income series from a quarterly to a monthly frequency. Additionally, `ggplot2` (version 3.5.1) is employed for data visualization and `dplyr` (version 1.1.4) is used for data manipulation.

4 Nowcasting and Forecasting German House Prices

4.1 Data

The dataset includes the quarterly German House Price Index (HPI) and national disposable income from Q2 2007 to Q4 2023, along with four monthly predictors from April 2007 to December 2023: the unemployment rate, residential building permits, the consumer price index and the housing loan interest rate. Additionally, it contains the daily closing prices of the DAX (Deutscher Aktien Index, the German stock index) from April 2, 2007, to December 29, 2023. The housing loan interest rate data is sourced from the Deutsche Bundesbank (www.bundesbank.de/) and the DAX historical data is obtained from Yahoo Finance (de.finance.yahoo.com). The remaining time series are taken from the databases of the Statistisches Bundesamt (www.destatis.de). Table 1 provides a summary of the variables.

The selection of variables is based on a review of relevant empirical literature. These variables cover both housing supply and demand factors, as well as alternative investment options to housing. The number of residential building permits, encompassing both new construction and renovations, serves as an indicator of housing supply as well as the enhancement of property value. Kishor (2023) conducts out-of-sample forecasting with bivariate and trivariate VAR models and finds that

Table 1: Summary of variables

Predictor	Code	Frequency	Publication lag (Months)
House price index	hpi	Quarterly	3
Dax close price	dax	Daily	0
Unemployment rate	unrate	Monthly	2
Residential building permits	rbp	Monthly	2
Consumer price index	cpi	Monthly	1
Housing loan interest rate	hl	Monthly	2
National disposable income*	din	Quarterly	3

Note: The table describes the time series used in the forecasting exercise. The dataset contains two quarterly, five monthly and one daily series over the sample period from April 2007 to December 2023 at quarterly, monthly and daily frequencies. A linear transformation (interpolation) is used to convert the quarterly national disposable income (*din*) series to monthly. Stationarity of the series is achieved by first differencing and all series except the DAX close price are seasonally adjusted. Refer to Section 4.1 for details on the publication lag.

the monthly supply of new homes is the most significant predictor of house price growth, leading to a substantial reduction in RMSFE compared to a univariate benchmark. Income is a key factor in determining housing demand. It is directly linked to purchasing power, which influences housing demand and subsequently affects house prices. Holly and Jones (1997) investigate various factors, such as real income, demography, interest rates and inflation, and conclude that real income is the most significant determinant of real house prices in the UK. Additionally, Hort (1998) provides empirical evidence in her long-term analysis that fluctuations in income significantly affect real house prices. Consumer prices are a measure of inflation. Inflation indirectly influences housing demand by affecting disposable income. As inflation rises, the cost of living increases, reducing the purchasing power for houses. The unemployment rate is an indirect factor for housing demand as well, affecting income levels. Higher unemployment results in income loss for many individuals, thereby reducing their ability to buy homes. In the study by Baffoe-Bonnie (1998), the impulse response functions derived from the VAR suggest that changes in employment significantly affect cycles in housing prices and the number of houses sold at both the national and regional levels. A lower interest rate reduces the costs of financing for purchasing a house. Therefore, much of the literature has addressed the impact of interest rates on housing prices. Hanck and Prüser (2022) use a Bayesian VAR to study that the German house price boom since 2010 can be explained by

falling interest rates. Their empirical analysis underscores the significant influence of interest rates, revealing that a substantial portion of the changes in house prices is driven by interest rate shocks. Berlemann and Freese (2013) employ VAR models to examine the response of house and rental prices in Switzerland to interest rate shocks. They find that increases in interest rates negatively impact both house and condominium prices, as well as rental prices. Building on previous studies about the impact of interest rates on house prices, I focus on the housing loan interest rate, which is expected to be more closely linked to house prices than total bank lending. In addition to the macroeconomic variables, I consider the DAX price as an indicator of the stock market. The performance of the DAX indicates the overall economic health, suggesting that during periods of economic growth, both the DAX and housing prices may increase concurrently. Conversely, stocks present an alternative investment option to real estate. When stock market returns are high, investors are inclined to direct more funds toward the stock market rather than real estate, which can lead to a decrease in real estate demand and potentially stabilize or lower housing prices.

The COVID-19 pandemic and subsequent actions, such as lockdowns, led to a global economic crisis that significantly affected Germany as well. According to Statistisches Bundesamt (March 2022), by 2021 Q4, German GDP was 1.1% below pre-crisis levels and employment was 0.4% lower than in Q4 2019. Figure 1 reports the movements of variables of interest over time. The dashed line indicates March 2020, the time when the first COVID-19 pandemic lockdown took effect. Around this point, all variables show significant changes. The German HPI, which had been steadily increasing, surged sharply following the implementation of the first COVID-19 lockdown in March 2020, reaching its peak in Q2 2022. This surge can be attributed to the growing demand for home office space and the desire for better living environments, as people were compelled to spend most of their time at home. During the same period, the housing loan interest rate remained low, reaching its lowest point of 1.16% in December 2020. However, from May 2022 onwards, the housing loan interest rate started to climb significantly, peaking at 4.22% in November 2023, during which time the HPI also decreased considerably. The co-movement of these variables indicates that housing loan interest rates can impact house prices and potentially serve as a strong indicator for

explaining house price changes. The DAX close price declined sharply in March 2020 when the first lockdown took effect but subsequently surged, increasing by 59% by November 2021. The unemployment rate, which had shown a decreasing trend before the lockdown, rose afterward and then returned to pre-lockdown levels by early 2022. The number of residential building permits, previously showing seasonal patterns, demonstrated a different movement after the lockdown. It peaked in March 2021 and then declined sharply, falling 62% by December 2023. The CPI began to increase sharply from March 2022, peaking with a 17% rise by the end of 2023. National disposable income fell sharply after the lockdown but gradually recovered thereafter.

All series, except for the DAX close price, have been seasonally adjusted using the X13 seasonal adjustment method implemented by JDemetra+, the software officially recommended by the European Statistical System (ESS) and the European System of Central Banks. Stationarity is obtained by log-differencing the time series, yielding the growth rates of the series. In the following discussion, these log-differenced variables are referred to by the abbreviations presented in Table 1.

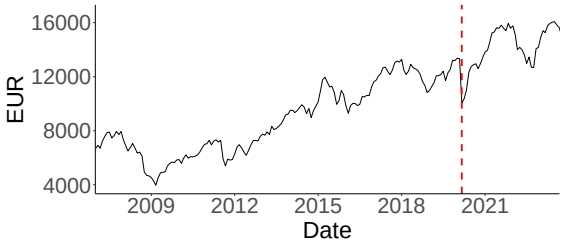
4.2 Forecasting with ragged-edge and pseudo real-time datasets

The publication of data by statistical agencies inherently involves a time lag, which can range from several weeks to several months due to the time required for data collection, processing and validation. This publication lag leads to missing values of variables at the time of analysis and results in ragged edges at the end of datasets.

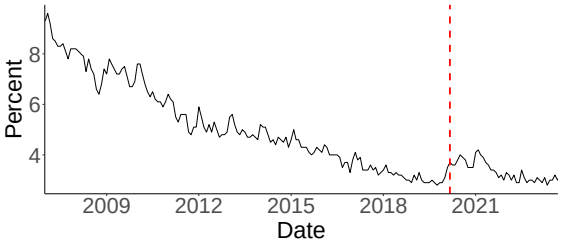
Recent empirical studies, such as Giannone, Reichlin and Small (2008) and Schumacher and Breitung (2008), highlight that replicating the varying availability of data, rather than using artificially balanced samples, can improve the accuracy of forecasts. Additionally, considering the practical issue that data of interest may not always be available in real-time, I use a dataset that includes final or most recently revised vintage data. Thus, I use data typically available from statistical agency databases within a pseudo-real-time framework, as described below.

In this framework, I assume that each forecast is computed at the end of the quarter and that the patterns of publication lags indicated in the column “Publication lag” in Table 1, remain the

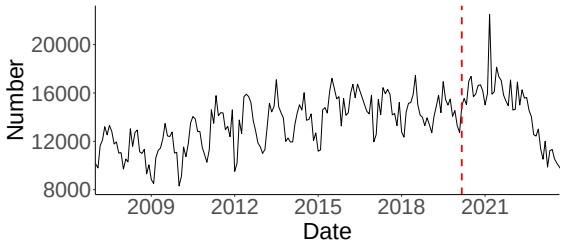
Figure 1: Time series overview



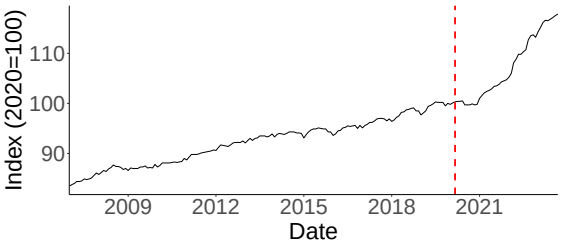
(a) DAX close price



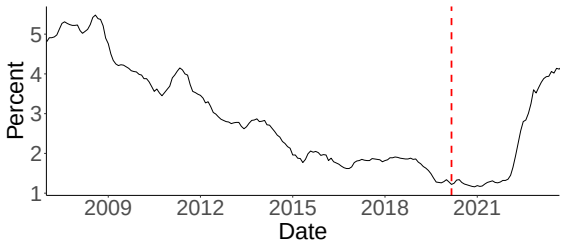
(b) Unemployment rate



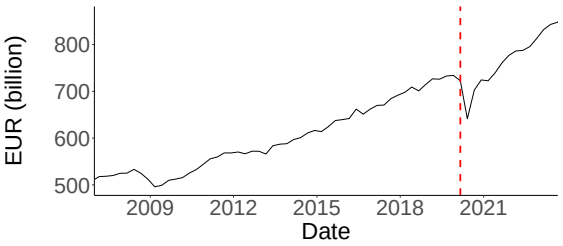
(c) Residential building permits



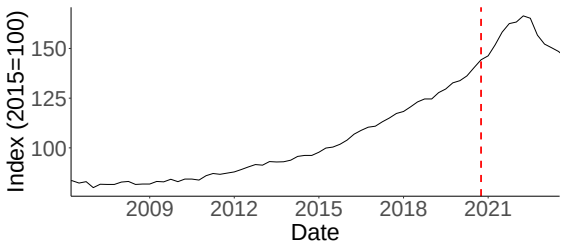
(d) Consumer price index



(e) Housing loan interest rate



(f) National disposable income



(g) House price index

Note: The dashed line indicates March 2020, the month when the first COVID-19 lockdown was enforced.

same over time. For instance, when forecasting *hpi* on March 31, 2007, *unrate* is available only up to January and *cpi* is available only up to February, resulting in two and one missing values, respectively. To replicate the publication lag of *hpi*, it is noted that the Federal Statistical Office of Germany publishes the HPI approximately 85 days after the end of the reporting quarter (Deutsche Bundesbank, 2020, October). This approach ensures that the forecasts reflect the actual data availability at the time of nowcasting and forecasting, thereby providing a more realistic assessment of model performance.

4.3 Nowcast and forecast design

This study evaluates the performance of mixed-frequency models for nowcasting and forecasting the German quarterly HPI. The full sample is divided into an evaluation sample and an estimation sample, which is recursively expanded over time. Each model is re-estimated as new data becomes available, allowing the coefficients to change over time. This approach enhances the models' ability to capture shifts in the underlying data generation process, which is particularly beneficial when an economic shock increases uncertainty.

To assess the impact of the economic shocks caused by the COVID-19 pandemic on the results, I consider two evaluation samples. Both start in 2018 Q1, but one ends in 2020 Q1, while the other extends to 2023 Q4. These different endpoints for the evaluation samples correspond to the period when the first lockdown was implemented in Germany during the COVID-19 pandemic, specifically on March 13 and March 17, 2020.

For each of the quarters in evaluation samples, nowcasts and forecasts are computed for forecast horizons $h = 0, 1, 2, 3, 4, 8$ quarters, where $h = 0$ represents forecasting the current quarter, commonly known as nowcasting. This range of horizons allows for a comprehensive performance assessment of mixed-frequency models from both short-term and long-term perspectives.

The MSFEs of mixed-frequency models for each predictor and horizon are divided by the MSFE of a benchmark, which is a recursively estimated autoregressive (AR) model, with the lag length recursively selected by the Bayesian Information Criterion (BIC). By computing the relative MSFE,

the MSFEs are adjusted to a common scale, making it easier to interpret and compare model performance. Additionally, comparing these models to a naive benchmark helps verify whether incorporating mixed-frequency data and accounting for data availability improves predictive accuracy. A relative MSFE less than 1 indicates that a mixed-frequency model outperforms the benchmark. Within a pseudo-real-time framework, the estimation samples of mixed-frequency models are recursively expanded over time, accounting for the patterns of missing values due to publication lags, as discussed in Section 4.2, while the benchmark uses a complete series of *hpi* without the missing values.

4.4 Model specification

MIDAS regressions are tightly parameterized and take a reduced form. The use of parsimonious distributed lag polynomials enables the straightforward incorporation of daily DAX index growth as higher-frequency explanatory variables for forecasting the quarterly HPI, see Section 3.1.1. Hence, for MIDAS and AR-MIDAS, DAX index growth retains its daily frequency, whereas it is aggregated to a monthly series for other models. The initial parameter set for NLS estimation of MIDAS and AR-MIDAS is determined by identifying the parameter pair that minimizes the residual sum of squares (RSS) from the sets $\theta_1 = \{-0.5, 0, 0.5\}$ and $\theta_2 = \{-0.01, -0.1, 0, -0.5, -1\}$, following Foroni, Marcellino and Schumacher (2015). The maximum number of lags chosen for MIDAS models is $K = 12$ months for monthly predictors and $K = 252$ days for a daily predictor. To address the issue of daily data having a different number of days in each quarter, it is assumed that each quarter comprises 64 days. In U-MIDAS, the maximum lag order of AR terms is set to $p = 4$. The BIC is recursively used to select the appropriate lag length for each evaluation sample as new data becomes available.

MF-VAR uses a lag length of $r = 4$ for monthly time series. The Gibbs sampler is used to generate a sample of 15,000 draws from the joint posterior distribution of the model parameters. As described in Section 3.2.4, the hyperparameters are selected by nowcasting performance across a range of values from the sets $\lambda_1 = \{0.01, 0.1, 0.5, 0.9\}$ and $\lambda_2 = \{0.1, 0.5, 1, 2, 4\}$. The steady-state

priors are specified 95% probability intervals for the variables, following previous literature, such as Ankargren, Bjellerup and Shahnazarian (2017), Ankargren, Unosson and Yang (2018) and Österholm (2008). In Figure 2, these intervals are visualized by green areas, with the dashed lines representing the prior means.

4.5 Empirical results

4.5.1 Single indicator models

The results for the first evaluation sample, shown in Table 2, reveal that MF-VAR provides outstanding nowcasts with relative MSFEs below one for all predictors except for *rbp*. In contrast, U-MIDAS, MIDAS and AR-MIDAS produce relative MSFEs above one for the nowcasting horizon $h = 0$ across all predictors. For short-term forecast horizons $h = \{1, 2\}$, the relative MSFEs of all models are consistently higher than one, indicating the difficulty of surpassing the benchmark in short-term forecasting. However, for *din* with horizons $h = \{3, 4\}$, the MSFEs of the MF-VAR and AR-MIDAS models decrease. For the longest forecast horizon $h = 8$, most MSFEs are below one, with AR-MIDAS achieving the lowest MSFEs across all predictors.

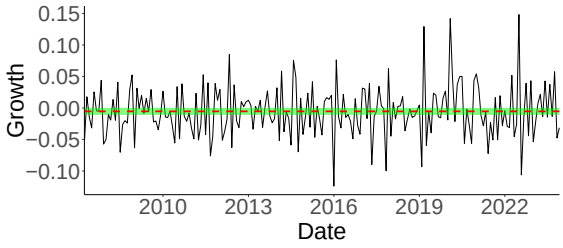
For the second evaluation sample, Table 3 presents that overall performance improves, with most relative MSFEs below one. For forecast horizons $h = \{1, 2, 3, 4, 8\}$, mixed-frequency models exhibit a trend where relative MSFEs decrease as the horizon increases, then slightly rise for the longest horizon $h = 8$. MF-VAR provides even better nowcasting performance compared to the first sample, while MIDAS models still hover around one. MF-VAR provides stable results across predictors for the second sample but produces relatively higher MSFEs compared to AR-MIDAS and U-MIDAS for horizons $h = \{1, 2, 3, 4, 8\}$. For short forecast horizons, MIDAS still produces uninformative results with MSFEs greater than or close to one. The distinction between AR-MIDAS and U-MIDAS is less pronounced. While AR-MIDAS generally exhibits the lowest MSFEs, as highlighted by the bold figures in the tables, U-MIDAS produces comparable MSFEs to AR-MIDAS or even outperforms AR-MIDAS for specific predictors and forecast horizons.

In order to assess the average performance of mixed-frequency models, the relative MSFEs of

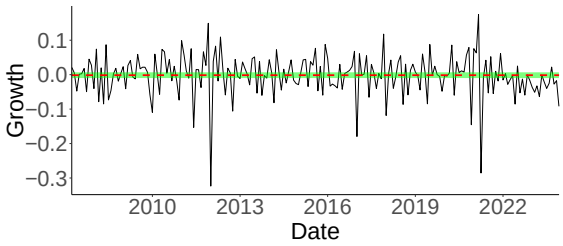
Figure 2: Steady-state priors



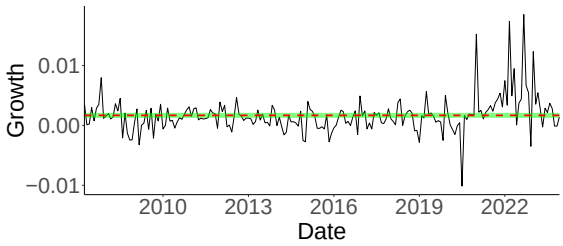
(a) DAX close price



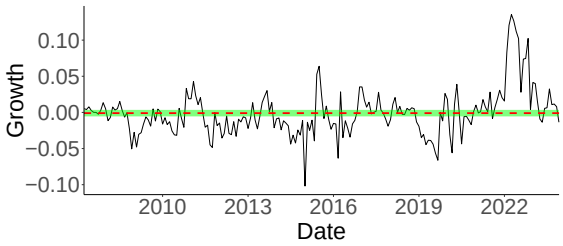
(b) Unemployment rate



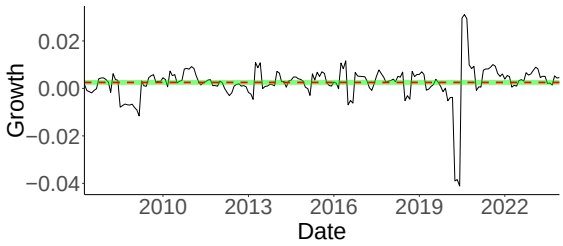
(c) Residential building permits



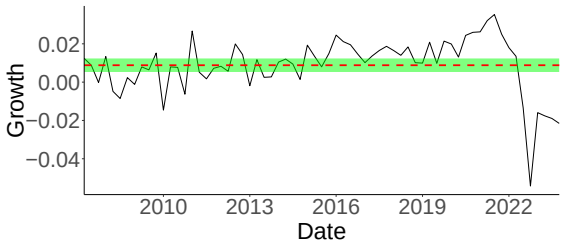
(d) Consumer price index



(e) Housing loan interest rate



(f) National disposable income



(g) House price index

Note: The green areas in the figures correspond to 95% prior probability intervals of the variables with the dashed line showing the prior mean.

Table 2: Results of individual mixed-frequency models for the first evaluation sample

predictor	Model	Horizon h					
		0	1	2	3	4	8
dax	MF-VAR	0.88	1.26	1.43	1.14	1.23	0.89
	U-MIDAS	1.56	1.57	1.59	1.21	1.27	1.69
	MIDAS	1.80	1.65	1.67	1.40	1.59	1.58
	AR-MIDAS	1.18	1.39	1.43	1.29	1.26	0.76
unrate	MF-VAR	0.89	1.55	1.62	1.11	1.30	1.00
	U-MIDAS	3.00	2.74	2.77	2.48	2.59	1.39
	MIDAS	3.51	2.73	2.82	2.35	2.45	1.80
	AR-MIDAS	2.19	2.38	2.38	1.95	2.04	**0.38**
rbp	MF-VAR	1.06	1.69	1.64	1.11	1.31	0.99
	U-MIDAS	1.37	1.27	1.24	1.00	1.04	0.70
	MIDAS	1.62	1.56	1.42	1.17	1.13	1.10
	AR-MIDAS	1.35	1.20	1.18	0.98	1.01	0.77
cpi	MF-VAR	0.84	1.58	1.47	1.22	1.35	0.92
	U-MIDAS	1.42	1.44	1.44	1.11	1.16	0.74
	MIDAS	1.59	1.47	1.40	1.34	1.32	1.06
	AR-MIDAS	1.37	1.29	1.33	1.05	1.32	0.67
hl	MF-VAR	0.89	1.40	1.42	1.15	1.21	0.92
	U-MIDAS	1.42	1.42	1.41	1.07	1.12	0.77
	MIDAS	1.58	1.56	1.56	1.12	1.16	0.87
	AR-MIDAS	1.39	1.33	1.41	1.02	1.08	0.73
din	MF-VAR	**0.82**	1.35	1.42	1.14	1.23	0.91
	U-MIDAS	1.32	1.31	1.32	1.05	1.09	0.86
	MIDAS	1.11	**1.05**	1.03	0.83	0.87	0.72
	AR-MIDAS	1.18	1.19	**1.00**	**0.82**	**0.80**	0.53

Note: The table reports the relative MSFEs of mixed-frequency models, computed by dividing the MSFE of each model by the MSFE of the AR benchmark for the evaluation sample ending in March 2020 (before the economic crisis caused by the COVID-19 pandemic). The first two columns in the table provide the predictor name and model type. The numbers in bold indicate the best performance for each horizon. For the meanings of the abbreviations of the predictors, see Table 1.

Table 3: Results of individual mixed-frequency models for the second evaluation sample

predictor	Model	Horizon h					
		0	1	2	3	4	8
dax	MF-VAR	**0.60**	0.82	0.86	0.71	0.71	0.81
	U-MIDAS	1.01	0.69	0.60	0.47	0.46	0.57
	MIDAS	1.70	1.34	1.36	0.85	0.82	0.83
	AR-MIDAS	1.02	0.67	0.65	0.49	0.47	0.54
unrate	MF-VAR	0.67	0.91	0.86	0.74	0.76	0.81
	U-MIDAS	0.99	0.70	0.61	0.49	0.48	0.60
	MIDAS	1.86	1.24	1.11	0.86	0.86	1.02
	AR-MIDAS	1.14	0.71	0.58	0.52	0.47	0.60
rbp	MF-VAR	0.62	0.95	0.88	0.72	0.72	0.81
	U-MIDAS	1.03	0.69	0.60	0.48	0.46	0.56
	MIDAS	1.20	0.93	0.81	0.64	0.58	0.65
	AR-MIDAS	0.92	**0.65**	**0.55**	**0.44**	**0.43**	**0.50**
cpi	MF-VAR	0.81	0.92	0.94	0.75	0.76	0.81
	U-MIDAS	1.17	0.84	0.59	0.47	0.47	0.56
	MIDAS	1.47	1.15	1.23	0.84	0.75	0.61
	AR-MIDAS	0.99	0.79	0.69	0.51	0.43	0.56
hl	MF-VAR	0.67	0.89	0.84	0.70	0.72	0.81
	U-MIDAS	0.96	0.67	0.58	0.46	0.44	0.53
	MIDAS	1.26	1.13	1.07	0.88	0.82	0.75
	AR-MIDAS	1.02	0.70	0.59	0.47	0.44	**0.50**
din	MF-VAR	0.71	0.89	0.87	0.73	0.73	0.81
	U-MIDAS	1.09	0.72	0.63	0.50	0.48	0.58
	MIDAS	1.97	1.33	1.27	0.95	0.87	0.94
	AR-MIDAS	1.21	0.71	0.65	0.57	0.44	0.64

Note: The table reports the relative MSFEs of mixed-frequency models, computed by dividing the MSFE of each model by the MSFE of the AR benchmark for the evaluation sample ending in December 2023 (with the economic crisis caused by the COVID-19 pandemic). The first two columns in the table provide the predictor name and model type. The numbers in bold indicate the best performance for each horizon. For the meanings of the abbreviations of the particular predictors, see Table 1.

each mixed-frequency model are averaged across predictors for each forecast horizon h. Table 4 depicts the average performance of these models separately for the evaluation samples ending before and after the economic crisis caused by the COVID-19 pandemic. For the first evaluation sample, the MSFEs of all models except for that of the eight-quarter-ahead forecast of AR-MIDAS are greater than or close to one across all horizons. MF-VAR and AR-MIDAS produce similar average MSFEs for forecast horizons $h = \{1,2,3,4\}$, while MF-VAR and AR-MIDAS yield the lowest MSFE for $h = 0$ and $h = 8$, respectively. The results for the second evaluation sample indicate that U-MIDAS and AR-MIDAS perform very similarly, demonstrating low MSFEs for forecast horizons $h = \{1,2,3,4,8\}$. Additionally, the average nowcasting performance of MF-VAR is superior to MIDAS models.

Table 4: Average relative MSFE of the mixed-frequency model to an AR benchmark

Model	Horizon h					
	0	1	2	3	4	8
Before the economic crisis caused by the COVID-19 pandemic						
MF-VAR	**0.90**	1.47	1.50	**1.14**	1.27	0.94
U-MIDAS	1.68	1.62	1.63	1.32	1.38	1.02
MIDAS	1.87	1.67	1.65	1.37	1.42	1.19
AR-MIDAS	1.44	**1.46**	**1.46**	1.18	**1.25**	**0.64**
With the economic crisis caused by the COVID-19 pandemic						
MF-VAR	**0.68**	0.90	0.88	0.72	0.73	0.81
U-MIDAS	1.04	0.72	**0.60**	**0.48**	0.46	0.57
MIDAS	1.58	1.19	1.14	0.84	0.78	0.80
AR-MIDAS	1.05	**0.70**	0.62	0.50	**0.45**	**0.56**

Note: The table reports the average performance of mixed-frequency models. The average MSFEs in the table are obtained as follows. First, the relative MSFEs of each mixed-frequency model are calculated for each predictor and forecast horizon. Second, the relative MSFEs are averaged across predictors. The table is divided into two parts. The top part presents the results for the evaluation sample ending in March 2020 (before the economic crisis caused by the COVID-19 pandemic), while the bottom part displays the results for the evaluation sample ending in December 2023 (with the economic crisis caused by COVID-19 pandemic). The numbers in bold indicate the best performance for each horizon within each evaluation sample period.

From the results of single predictor models and their average performance, it is confirmed that the ranking of U-MIDAS and AR-MIDAS is unclear. Hence, I visualize the ratio between the MSFEs

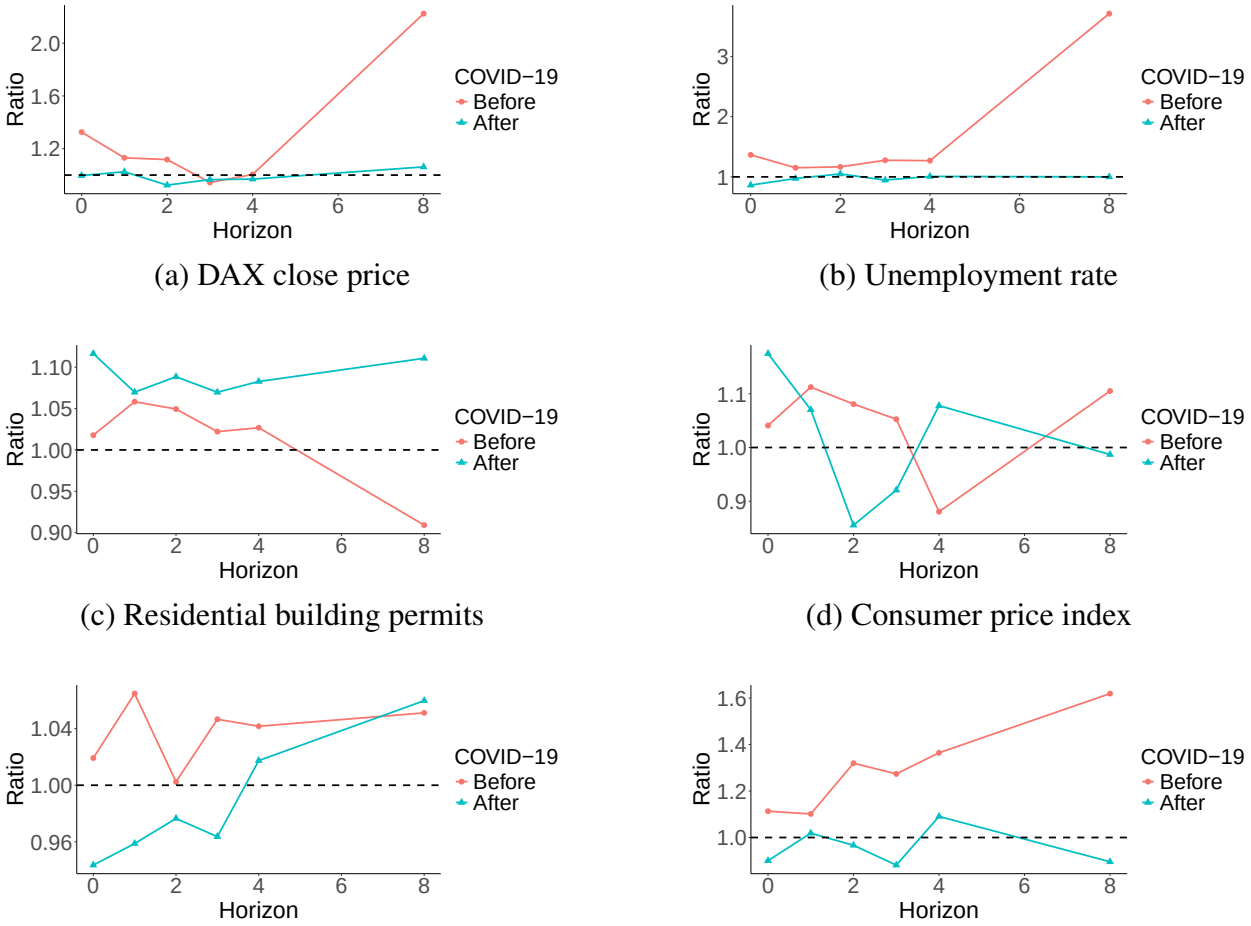
of U-MIDAS and AR-MIDAS for each predictor and conduct the Diebold-Mariano test to examine whether the difference in their nowcast and forecast accuracy is statistically significant. The null hypothesis for the test posits that "There is no difference in accuracy between the forecasts of the competing models."

Figure 3 displays the relative performance of these models for two evaluation samples ending before and after the COVID-19 pandemic. The ratio is calculated by dividing the MSFE of U-MIDAS by that of AR-MIDAS for each predictor and forecast horizon. The dashed line in the figure indicates that the ratio is equal to one. A value lower than one indicates that U-MIDAS outperforms AR-MIDAS. According to the results for the first evaluation sample, represented by the red lines, the ratios of the two models rapidly change for the forecast horizon $h = 8$, except for *hl*. For other horizons, the ratios remain relatively stable. The corresponding units on the y-axis of the figure reflect minor variations across horizons. The changes in ratios for the second evaluation sample are even more stable, indicating that U-MIDAS and AR-MIDAS produce similarly accurate nowcasts and forecasts across predictors and horizons. Table 5 presents the results of the Diebold-Mariano test. The test is conducted at a 95% confidence level, where p-values under 0.05 signify that the differences in forecast accuracy between U-MIDAS and AR-MIDAS are statistically significant. The test results for the second evaluation sample suggest that both models exhibit similar forecast accuracy across most horizons and predictors, with the exception of *rbp* at horizon $h = 8$. This highlights U-MIDAS as a strong competitor to AR-MIDAS, offering practical advantages due to its computational simplicity. U-MIDAS can be efficiently estimated using OLS, without the need for restrictions on lag polynomials or AR components, resulting in a much faster estimation process.

Overall, the analysis of mixed-frequency models for individual predictors reveals that these models significantly outperform the benchmark during periods of economic uncertainty. Appendix 5 shows that the absolute MSFEs of all models, including the benchmark, are significantly higher for the second evaluation sample compared to the first, reflecting the increased volatility during this period. However, the relative MSFEs of the models compared to the benchmark are markedly lower for the second sample. This suggests that mixed-frequency models produce more accurate and

stable nowcasts and forecasts under economically unstable conditions. The relative performance of mixed-frequency models varies depending on the predictor and the forecast horizon. MF-VAR is particularly effective for nowcasting, while AR-MIDAS models are advantageous for forecast horizons of $h = \{1,2,3,4,8\}$ quarters. The outstanding performance of AR-MIDAS compared to the basic MIDAS highlights the importance of including the AR component, which significantly enhances forecast accuracy. Lastly, the comparative analysis of U-MIDAS and AR-MIDAS indicates that U-MIDAS is a valuable alternative to AR-MIDAS.

Figure 3: Relative results of U-MIDAS to AR-MIDAS for single predictors



(a) DAX close price

(b) Unemployment rate

(c) Residential building permits

(d) Consumer price index

(e) Housing loan interest rate

(f) National disposable income

Note: This figure shows the ratio between MSFEs of U-MIDAS and AR-MIDAS for each predictor. A value lower than one indicates that the MSFE of U-MIDAS is lower than that of AR-MIDAS, meaning that U-MIDAS performs better than AR-MIDAS. The red line represents the results for the evaluation sample ending before the COVID-19 pandemic, while the blue line represents the results for the evaluation sample ending after the pandemic.

Table 5: The Diebold-Mariano test for the forecast accuracy of U-MIDAS and AR-MIDAS

predictor	Horizon h					
	0	1	2	3	4	8
dax	0.930	0.325	0.372	0.238	0.511	0.117
unrate	0.235	0.730	0.540	0.575	0.962	0.999
rbp	0.105	0.164	0.144	0.485	0.343	0.303
cpi	0.434	0.706	0.179	0.195	0.244	0.754
hl	0.448	0.609	0.644	0.561	0.532	0.042
din	0.437	0.137	0.467	0.338	0.294	**0.007**

Note: This table presents the p-values from the Diebold-Mariano test, which compares the forecast accuracy of the U-MIDAS and AR-MIDAS models for the second evaluation sample ending in December 2023 (with the economic crisis caused by the COVID-19 pandemic). The null hypothesis posits that there is no difference in forecast accuracy between the competing models. The test is conducted at a 95% confidence level, with p-values lower than 0.05 indicating statistically significant differences in forecast accuracy. For the meanings of the abbreviations of the predictors, see Table 1.

4.6 Combined predictor models

Empirical studies have often shown that combining forecasts yields better average results than individual models. Forecast combination can help address issues such as model misspecification, parameter instability and measurement errors in datasets associated with individual models, as discussed by Timmermann (2006). In this subsection, the nowcast and forecast accuracy of mixed-frequency models is evaluated for combined series consisting of all predictors used for single predictor models.

The evaluation process is conducted in two stages. Initially, I compare the relative MSFEs of the pooled mixed-frequency models to the AR benchmark. Subsequently, I analyze the percentiles of these computed relative MSFEs within the empirical distribution of all MSFEs for individual predictors in a given class of models. The results are detailed in Table 6 and Table 7, respectively. In Table 6, the MSFEs of the combined predictor models for the first evaluation sample are significantly higher than one, except for the MSFEs of MF-VAR at horizons $h = \{0, 3\}$. In contrast, the results for the second evaluation sample demonstrate that the mixed-frequency models generally outperform the benchmark, except for MIDAS. Consistent with the results for single predictor models, the mixed-frequency models outperform the benchmark when the evaluation sample

includes the crisis period.

Across both evaluation samples, MF-VAR demonstrates the best nowcasting performance at $h = 0$. The rank for the second sample changes over forecast horizons. MF-VAR, U-MIDAS and AR-MIDAS each produce the lowest MSFE at specific horizons. In contrast, MIDAS yields uninformative results across both samples, with all relative MSFEs exceeding one. Notably, U-MIDAS produces contrasting results across the evaluation samples. For the first sample, it generates much higher MSFEs than any other model, while for the second sample, its accuracy improves significantly, producing the lowest MSFEs for horizons $h = 1, 8$. This unstable performance of U-MIDAS is likely due to its linearity. When the number of predictors and the number of lags are too large, U-MIDAS can suffer significantly from overparametrization.

Table 6: Results of pooled models relative to an AR benchmark

Model	Horizon h					
	0	1	2	3	4	8
Before the economic crisis caused by the COVID-19 pandemic						
MF-VAR	**0.96**	**1.62**	**1.26**	**0.86**	**1.22**	**1.14**
U-MIDAS	3.84	2.71	2.60	2.48	2.46	6.07
MIDAS	2.26	2.47	1.84	1.44	1.83	2.57
AR-MIDAS	2.00	1.86	2.07	1.46	1.58	2.27
With the economic crisis caused by the COVID-19 pandemic						
MF-VAR	**0.64**	0.86	**0.83**	0.70	0.71	0.81
U-MIDAS	1.19	**0.80**	0.71	0.58	0.63	**0.60**
MIDAS	1.79	1.79	1.65	1.31	1.13	1.20
AR-MIDAS	1.34	0.89	0.67	**0.50**	**0.46**	0.62

Note: The table reports the relative MSFEs of pooled mixed-frequency models, computed by dividing the MSFE of each pooled model by the MSFE of the AR benchmark. The table is divided into two parts. The top part presents the results for the evaluation sample ending in March 2020 (before the economic crisis caused by the COVID-19 pandemic), while the bottom part displays the results for the evaluation sample ending in December 2023 (with the economic crisis caused by the COVID-19 pandemic). The first two columns in the table provide the model type. The numbers in bold indicate the best performance for each horizon.

According to the results in Table 7, all MIDAS models produce uninformative forecasts across both evaluation samples for all horizons. The MIDAS models consistently show high percentiles, indicating that the majority of MSFEs from individual MIDAS models are lower than those from

multivariate MIDAS models. In contrast, MF-VAR shows competitive performance. In terms of the first evaluation sample, MF-VAR has percentiles from 0 to 0.17 for horizon $h = \{2,3\}$, that is, only one or none of the individual MF-VAR models have lower MSFEs compared to the multivariate MF-VAR. Furthermore, for the second evaluation sample, MF-VAR maintains low percentiles across horizon $h = \{0,1,2,3,4\}$. These low percentiles indicate that the pooled MF-VAR is a useful alternative to forecasting with individual MF-VAR models.

Table 7: Percentiles of the MSFEs of pooled mixed-frequency models

Model	Horizon h					
	0	1	2	3	4	8
Before the economic crisis caused by the COVID-19 pandemic						
MF-VAR	0.83	0.83	0.00	0.00	0.17	1.00
U-MIDAS	1.00	0.83	0.83	1.00	0.83	1.00
MIDAS	0.83	0.83	0.83	0.83	0.83	1.00
AR-MIDAS	0.83	0.83	0.83	0.83	0.83	1.00
With the economic crisis caused by the COVID-19 pandemic						
MF-VAR	0.33	0.17	0.00	0.17	0.00	1.00
U-MIDAS	1.00	0.83	1.00	1.00	1.00	1.00
MIDAS	0.67	1.00	1.00	1.00	1.00	1.00
AR-MIDAS	1.00	1.00	0.83	0.50	0.67	0.83

Note: The table reports the percentiles of MSFEs of pooled forecasts in Table 6 in the empirical distribution of all MSFEs of the single predictor models in Table 2 and 3 within a given class of models. The table is divided into two parts. The top part presents the results for the evaluation sample ending March 2020 (before the economic crisis caused by the COVID-19 pandemic), while the bottom part displays the results for the evaluation sample ending December 2023 (with the economic crisis caused by the COVID-19 pandemic). The first two columns in the table provide the model type.

5 Conclusion

This study compares state-of-the-art mixed-frequency models, including MIDAS, AR-MIDAS, U-MIDAS and MF-VAR in nowcasting and forecasting German house prices. These models account for different sampling frequencies and missing values due to publication lags, allowing the use of the most recent daily or monthly observations of predictors to provide timely updates of the German

HPI.

MIDAS, AR-MIDAS and U-MIDAS are single-equation models that regress quarterly German HPI on higher-frequency predictors, serving as direct multi-step forecasting tools. MIDAS and AR-MIDAS use parsimonious distributed lag polynomials to prevent parameter proliferation, while U-MIDAS, with no restriction on lag polynomials, offers computational simplicity via OLS estimation. Conversely, MF-VAR, a system approach, jointly models the dynamics between variables and operates as an iterative device. By incorporating Bayesian methods, MF-VAR shrinks the model, reduces parameter uncertainty and improves forecast accuracy. The Gibbs sampler is used to draw parameters and latent observations, providing monthly German HPI from the given data.

Ranking these models based solely on theoretical perspectives is not feasible; hence, their relative performance must be evaluated in specific empirical applications. In this paper, the models are assessed based on their ability to nowcast and forecast German house prices. The accuracy of nowcasts and forecasts for quarterly German HPI are compared using relative MSFE to a benchmark recursively estimated AR model. Two evaluation samples, ending before and after the economic crisis caused by the COVID-19 pandemic, are used to consider the impact of the crisis on the results. The main findings indicate that mixed-frequency models outperform the benchmark under unstable economic conditions. MF-VAR demonstrates exceptional nowcasting accuracy compared to MIDAS models, while AR-MIDAS produces the best forecasts for horizons ranging from one to eight quarters ahead. U-MIDAS proves to be a strong alternative to AR-MIDAS, providing competitively accurate forecasts with computational simplicity. Results of the mixed-frequency models for the combined dataset show that incorporating more information from multiple predictors is a good strategy for MF-VAR, whereas the MIDAS models for the combined dataset result in poorer nowcasting and forecasting accuracy compared to the model for a single predictor.

The empirical results of this study present findings that differ from previous research comparing mixed-frequency models. For instance, Kuzin, Marcellino and Schumacher (2011) find that AR-MIDAS outperforms MIDAS and MF-VAR at short forecast horizons (up to three months), while MF-VAR performs better at longer horizons (up to nine months). In contrast, Foroni and Marcellino

(2014) find that AR-MIDAS consistently outperforms other models, particularly at very short horizons, where MF-VAR shows poorer performance. However, this study shows that MF-VAR produces the most accurate nowcasts, while AR-MIDAS provides the most accurate forecasts for one- to eight-quarter horizons. The outstanding nowcasting performance of MF-VAR in this study may be attributed to the incorporation of prior information, such as hyperparameters, the prior mean of AR(1) parameters and the prior distribution of the unconditional mean. These differences underscore that the relative performance of models is highly dependent on specific application settings, such as the selection of the target variable, predictors, horizon and sample periods. Additionally, while previous studies find that incorporating more variables in a model generally enhances accuracy, the results of this study reveal that although this strategy improves the performance of MF-VAR, it diminishes the accuracy of MIDAS models. This discrepancy may stem from differences in how the series are combined. In their studies, predictors are pooled using a weighting scheme, whereas in this study, the series of predictors are combined without such constraints. Consequently, future research may benefit from exploring appropriate weighting schemes, which could potentially further enhance the performance of these models.

The results of the empirical application in this study present that relative MSFE with the benchmark model tends to be lower at longer horizons. This is attributed to the inclusion of economic variables, which allow the model to better adapt to changing economic conditions. However, a primary motivation for using mixed-frequency models is their capacity to provide timely forecasts for the near future, enabling proactive decision-making. Hence, future research could consider alternative benchmark models, such as a Bayesian VAR for quarterly series and conduct a thorough investigation into whether high-frequency variables enhance model accuracy, particularly at shorter horizons.

Finally, future research could explore alternative specifications for mixed-frequency models. For example, MIDAS weighting schemes, such as the Beta scheme proposed by Ghysels, Sinko and Valkanov (2007), and the Step Weighting functions applied by Armesto, Engemann and Owyang (2010). A potential variant for the steady-state prior is its hierarchical specification proposed by Louzis (2019). This approach is based on the hierarchical normal-gamma shrinkage prior proposed

by Brown and Griffin (2010). The advantage of this approach is that researchers only need to specify prior means of the steady-state parameters, while the prior variances are determined by hyperpriors. A more flexible model could also be of interest. The common stochastic volatility model proposed by Carriero, Clark and Marcellino (2016) incorporates a factor that allows for time-varying error covariance.

Appendix

Table 8: MSFEs for the first evaluation sample

predictor	Model	Horizon h					
		0	1	2	3	4	8
	AR	0.000038	0.000039	0.000044	0.000055	0.000064	0.000122
dax	MF-VAR	0.000033	0.000049	0.000062	0.000063	0.000078	0.000109
	U-MIDAS	0.000059	0.000062	0.000070	0.000067	0.000081	0.000206
	MIDAS	0.000068	0.000065	0.000073	0.000078	0.000101	0.000193
	AR-MIDAS	0.000044	0.000054	0.000062	0.000071	0.000080	0.000093
unrate	MF-VAR	0.000033	0.000061	0.000071	0.000062	0.000083	0.000122
	U-MIDAS	0.000113	0.000108	0.000121	0.000138	0.000165	0.000171
	MIDAS	0.000132	0.000107	0.000123	0.000130	0.000156	0.000221
	AR-MIDAS	0.000082	0.000094	0.000104	0.000108	0.000130	0.000046
rbp	MF-VAR	0.000040	0.000066	0.000072	0.000061	0.000083	0.000121
	U-MIDAS	0.000052	0.000050	0.000054	0.000055	0.000066	0.000085
	MIDAS	0.000061	0.000061	0.000062	0.000065	0.000072	0.000135
	AR-MIDAS	0.000051	0.000047	0.000052	0.000054	0.000064	0.000094
cpi	MF-VAR	0.000032	0.000062	0.000064	0.000068	0.000086	0.000112
	U-MIDAS	0.000053	0.000057	0.000063	0.000061	0.000074	0.000090
	MIDAS	0.000060	0.000058	0.000061	0.000074	0.000084	0.000130
	AR-MIDAS	0.000051	0.000051	0.000058	0.000058	0.000084	0.000082
hl	MF-VAR	0.000034	0.000055	0.000062	0.000064	0.000077	0.000112
	U-MIDAS	0.000053	0.000056	0.000062	0.000059	0.000071	0.000094
	MIDAS	0.000059	0.000061	0.000068	0.000062	0.000074	0.000116
	AR-MIDAS	0.000052	0.000052	0.000062	0.000057	0.000068	0.000089
din	MF-VAR	0.000031	0.000053	0.000062	0.000063	0.000078	0.000112
	U-MIDAS	0.000049	0.000051	0.000058	0.000058	0.000070	0.000105
	MIDAS	0.000042	0.000041	0.000045	0.000046	0.000056	0.000088
	AR-MIDAS	0.000044	0.000047	0.000044	0.000046	0.000051	0.000065

Table 9: MSFEs for the second evaluation sample

predictor	Model	Horizon h					
		0	1	2	3	4	8
	AR	0.000250	0.000374	0.000468	0.000636	0.000723	0.000802
dax	MF-VAR	0.000150	0.000307	0.000404	0.000454	0.000516	0.000648
	U-MIDAS	0.000254	0.000259	0.000282	0.000302	0.000333	0.000461
	MIDAS	0.000424	0.000500	0.000638	0.000542	0.000595	0.000665
	AR-MIDAS	0.000255	0.000252	0.000306	0.000313	0.000343	0.000434
unrate	MF-VAR	0.000168	0.000341	0.000401	0.000469	0.000547	0.000653
	U-MIDAS	0.000247	0.000260	0.000284	0.000312	0.000346	0.000481
	MIDAS	0.000465	0.000466	0.000521	0.000546	0.000622	0.000821
	AR-MIDAS	0.000286	0.000267	0.000271	0.000329	0.000343	0.000480
rbp	MF-VAR	0.000155	0.000357	0.000413	0.000456	0.000521	0.000651
	U-MIDAS	0.000258	0.000260	0.000281	0.000303	0.000334	0.000446
	MIDAS	0.000299	0.000350	0.000378	0.000407	0.000417	0.000521
	AR-MIDAS	0.000231	0.000243	0.000258	0.000283	0.000308	0.000402
cpi	MF-VAR	0.000202	0.000343	0.000441	0.000480	0.000547	0.000650
	U-MIDAS	0.000291	0.000316	0.000277	0.000300	0.000338	0.000446
	MIDAS	0.000367	0.000432	0.000575	0.000534	0.000544	0.000493
	AR-MIDAS	0.000248	0.000295	0.000323	0.000326	0.000313	0.000452
hl	MF-VAR	0.000168	0.000334	0.000392	0.000444	0.000520	0.000653
	U-MIDAS	0.000291	0.000316	0.000277	0.000300	0.000338	0.000446
	MIDAS	0.000424	0.000500	0.000638	0.000542	0.000595	0.000665
	AR-MIDAS	0.000255	0.000252	0.000306	0.000313	0.000343	0.000434
din	MF-VAR	0.000178	0.000333	0.000409	0.000461	0.000531	0.000647
	U-MIDAS	0.000273	0.000271	0.000296	0.000320	0.000350	0.000462
	MIDAS	0.000492	0.000496	0.000595	0.000606	0.000630	0.000757
	AR-MIDAS	0.000303	0.000266	0.000306	0.000363	0.000321	0.000516

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